

On four-sheeted polynomial mappings of $\mathbb{C}^2$. II. The general case

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Abstract. In this paper we prove that there are no four-sheeted polynomial mappings of $\mathbb{C}^2$ into itself whose Jacobian is a non-zero constant.

Introduction

Let $\tilde{\mathbb{C}}^2$ and $\mathbb{C}^2$ be two copies of the complex plane, and let $f : \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$ be a polynomial mapping whose Jacobian is a non-zero constant. The well-known Jacobi conjecture states that f is polynomially invertible (see [1], [2].)

The mapping f is polynomially invertible if and only if a generic point in $\mathbb{C}^2$ has exactly one pre-image. We define the *topological degree of a mapping* to be the number of pre-images of a generic point. It is known that there are no polynomial mappings of degree two or three whose Jacobian is a non-zero constant. The case of degree two is elementary, and the case of degree three was considered in [3]. The main result of this paper is as follows.

Theorem. *Let $f : \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$ be a polynomial mapping of topological degree four. Then the Jacobian of f cannot be a non-zero constant.*

This paper is a continuation of [4], where the case when f has one dicritical component was proved to be impossible. Here we consider the remaining cases. The proof of the theorem uses the notation and assertions in [4], § 1.

The author is deeply grateful to A. G. Vitushkin and S. Yu. Orevkov for useful discussions and for their help in preparing this paper.

§ 1. Properties of dicritical components and their resolutions at infinity

We now begin the proof of the theorem. Assume the opposite: there is a polynomial mapping f of topological degree 4 whose Jacobian is a non-zero constant.

Using finitely many σ -processes at infinity, we can extend f to a regular rational mapping $F : \tilde{X} \rightarrow X$, where $\tilde{X}$, X are non-singular compact complex varieties containing $\tilde{\mathbb{C}}^2$ and $\mathbb{C}^2$ respectively as open subsets. The curves $\tilde{L} := \tilde{X} \setminus \tilde{\mathbb{C}}^2$ and $L := X \setminus \mathbb{C}^2$ consist of finitely many non-singular rational components with transversal

This work was supported by the Russian Foundation for Basic Research (grant no. 96-01-01218).

AMS 1991 Mathematics Subject Classification. 13B10, 14E05.

and at most pairwise intersections. As in [4], we put $\tilde{L}_\infty := F^{-1}(L)$. An irreducible component $\tilde{l} \subset \tilde{L} - \tilde{L}_\infty$ is called a *dicritical component* of F if F is non-constant on $\tilde{l}$. Without loss of generality, we suppose that the following proposition is true.

Proposition 1.1. *F satisfies the minimality condition: there are no (-1) -curves $\tilde{c} \subset \tilde{X}$ (that is, non-singular rational curves with $\tilde{c}^2 = -1$) such that $F|_{\tilde{c}}$ is constant.*

Proposition 1.2. 1) *F has two dicritical components $\tilde{g}_1$ and $\tilde{g}_2$. The map F is locally biholomorphic on $(\tilde{g}_1 \cup \tilde{g}_2) \setminus \tilde{L}_\infty$, and its orders of ramification along $\tilde{g}_1$ and $\tilde{g}_2$ are equal to one and two respectively, that is, we have $m(\tilde{g}_1) = n(\tilde{g}_1) = 1$, $m(\tilde{g}_2) = 1$, $n(\tilde{g}_2) = 2$. (The numbers m, n are defined in Definition 3 of [4].)*

2) *F is non-constant on every connected component of $\tilde{L} - \tilde{L}_\infty$. That is, $\tilde{L} = \tilde{L}_\infty + \tilde{g}_1 + \tilde{g}_2$.*

Proof. 1) The number of dicritical components of F is greater than 1 (see [4]). If 1) is not valid, then Lemma 4.2 of [3] implies that F is an unbranched covering over $\mathbb{C}^2$ minus finitely many points. But this is impossible since the base is simply connected.

2) This follows from Proposition 1.1 and [3], Lemma 5.2.

As in [4], we use a tilde to denote vertices of $\tilde{L}$, and we remove the tilde to denote their images under F . For example, $a := F(\tilde{a})$. We put $g := g_1 \cup g_2$.

Proposition 1.3. *$\tilde{X}$ and X can be chosen to have the following properties.*

1) *The curve $\tilde{g}_i$ ($i = 1, 2$) intersects $\tilde{L}_\infty$ transversally at a unique point $\tilde{p}_i$.*

2) *The curves g_i ($i = 1, 2$) transversally intersect L . The splice diagram of g near L takes one of the forms depicted in Fig. 1.*

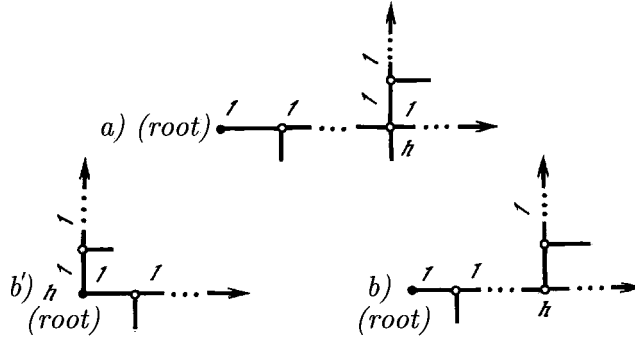


Figure 1

The right (resp. top) arrow corresponds to g_2 (resp. g_1). The graph L is the weighted part of $\Gamma_{L,g}$. For a nodal vertex $a \in L$, we denote by R_a, L_a, D_a, U_a the branches (possibly empty) of L at a in the directions Right, Left, Down, Up respectively. We put $U := U_h$. (The vertex h is defined in Fig. 1.) Then the following holds.

- a) If $a \in U$, then $\det U_a = 1$, $\det R_a > 1$. If $a \in L - U$ and $a \neq h$, then $\det R_a = 1$, $\det D_a > 1$, $\det L_a > 1$.

- b) In the case Fig. 1a), we have $\det U = 1$, $\det R_h = 1$, $\det D_h > 1$, $\det L_h > 1$.
- c) In the case Fig. 1b) with $\det U \neq 1$ we have $\det R_h = 1$, $\det L_h > 1$.
- d) $\det L = -1$.

Proof. Applying a polynomial automorphism of $\mathbb{C}^2$ (if necessary), we may assume the affine coordinates (x, y) in $\mathbb{C}^2$ to be such that neither $\deg_x g_2$ divides $\deg_y g_2$, nor $\deg_y g_2$ divides $\deg_x g_2$. Note that $g_1 \neq g_2$. Indeed, if $g_1 = g_2$, then g_1 has no singular points by [3], Lemma 5.2 and Proposition 1.2, 1). Applying the Abhyankar–Moh theorem as in Lemma 5.3 of [3], we see that F has a ramification of order 4 over g_1 . This contradicts Proposition 1.2. Resolving the singularity of g_2 at infinity yields the graph L' depicted in Fig. 2 of [4], and L is obtained from L' by resolving the singularity of g_1 at the point $p := g_1 \cap L'$. Each of $\tilde{g}_1, \tilde{g}_2$ has one point at infinity by [3], Lemma 2.1. This proves 1). Each of g_1, g_2 also has one point at infinity. Hence p is uniquely defined. Let l' be the vertex of L' such that $p \in l'$. If l' is nodal, then L corresponds to Fig. 1a). If l' is the root of L' , then L corresponds to Fig. 1b'). In other cases, L corresponds to Fig. 1b). Assertions 2)a)–d) follow from [4], Lemma 3 and [4], Proposition 3.

Convention. The graphs depicted in Fig. 1a) are said to be of type 1, those in Fig. 1b) with $\det U \neq 1$ are said to be of type 2, and other graphs in Fig. 1 are said to be of type 3. We consider the graphs depicted in Fig. 1b') as a particular case of those in Fig. 1b) such that $\det U = \det L_h = 1$. For graphs of type 2, we put $D_h := U$. We do not supply the graphs mentioned in Definition 1 of [4] with any labels if they are regarded as subgraphs of $\tilde{L}$ or L (for example, (ab)). The graphs regarded as subgraphs of $\tilde{L}_\infty$ will be labelled by ∞ (for example, $(\tilde{a}\tilde{b})_\infty$).

Corollary 1.4. a) Suppose that $a \in L - U$, $b \in L - U - h$ are nodal vertices and b lies to the left of a . Then $\det(ab)$ and $\det D_a$ are coprime.

b) Suppose that L is of type 1 or 2, and the nodal vertex a lies to the right of h . Then $\det(ah)$ and $\det D_a$ are coprime.

Proof. This follows from Proposition 1.3 and [4], Lemma 2.

Lemma 1.5. Let $\tilde{a}$ be a vertex of $\tilde{L}_\infty$. If the subgraph $(\tilde{a}\tilde{g}_1)$ is linear, then $\text{Deg}(\tilde{a}\tilde{g}_1) = 1$. If the subgraph $(\tilde{a}\tilde{g}_2)$ is linear, then $\text{Deg}(\tilde{a}\tilde{g}_2) = 2$.

Proof. This follows from Propositions 1.2, 1), 1.3 and [4], Proposition 1.

§ 2. Auxiliary statements

The following lemma is proved similarly to [4], Lemma 12.

Lemma 2.1. Let $\tilde{K} = \sum k_{\tilde{s}} \tilde{s}$ be the representation of the canonical class of $\tilde{X}$ as a linear combination of irreducible components $\tilde{s}$ of the curve $\tilde{L}$. Then $k_{\tilde{g}_i} = i - 1$, $i = 1, 2$.

Throughout the paper, we denote the root of $\tilde{L}$ by $\tilde{v}$. We note that $\tilde{v} \in \tilde{L}_\infty$.

Definition 2.2. A nodal vertex $\tilde{q} \in \langle \tilde{g}_1 \tilde{v} \rangle$ (resp. $\langle \tilde{g}_2 \tilde{v} \rangle$) is said to be *essential* for $\tilde{g}_1$ (resp. $\tilde{g}_2$) if there is a connected component $\Lambda \subset \delta(\tilde{g}_1 \tilde{v})$ (resp. $\Lambda \subset \delta(\tilde{g}_2 \tilde{v})$) incident to $\tilde{q}$ such that $\det \Lambda > 1$. This component Λ is unique by [4], Lemma 3. We put $d(\tilde{q}) := \det \Lambda$. Otherwise the vertex $\tilde{q}$ is said to be *inessential* for $\tilde{g}_1$ (resp. $\tilde{g}_2$).

Vertices inessential for $\tilde{g}_1$ ($\tilde{g}_2$) contribute nothing to formula (3) in [4] for the coefficient $k_{\tilde{g}_1}$ ($k_{\tilde{g}_2}$). We omit the expressions “for $\tilde{g}_i$ ” ($i = 1, 2$) when the situation is clear from the context, for example, when $\tilde{v} \in \langle \tilde{g}_1 \tilde{g}_2 \rangle$.

Lemma 2.3. *Let $\tilde{q} \in \langle \tilde{g}_1 \tilde{v} \rangle$ be a nodal vertex, and suppose that $\det(\tilde{q}\tilde{g}_1) = \det \delta(\tilde{q}\tilde{g}_1) = 1$. Then $\tilde{q}$ is inessential for $\tilde{g}_1$.*

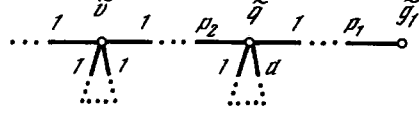


Figure 2

Proof. Assume the opposite. Put $d = d(\tilde{q})$, $p_1 = \det \text{br}_{\tilde{g}_1}(\tilde{v})$, $p_2 = \det \text{br}_{\tilde{q}}(\tilde{v})$. Then the splice diagram of $\tilde{L}$ is as in Fig. 2. Using formula (2) in [4] for the determinant of $(\tilde{g}_1 \tilde{q})$ regarded as a subgraph of $\tilde{L}$, we get

$$-1 = p_1 - dp_2. \quad (2.1)$$

It follows from [4], (3) and Lemma 2.1 that

$$0 = k_{\tilde{g}_1} = -2 - p_1 + (d-1)(p_2-1) + d\Sigma, \quad (2.2)$$

where Σ is the remainder in [4], (3). Substituting (2.1) in (2.2), we obtain that p_2 is divisible by $d \neq 1$. This contradicts [4], Lemma 2.

Lemma 2.4. *Suppose that $\tilde{q}$ satisfies the hypotheses of Lemma 2.3, and that $\det \text{br}_{\tilde{s}}(\tilde{v}) = 1$ for any vertex $\tilde{s} \in \langle \tilde{q}\tilde{v} \rangle$ which is essential for $\tilde{g}_1$. (This condition holds automatically if $(\tilde{v}\tilde{q})$ is linear.) Then $\det \text{br}_{\tilde{q}}(\tilde{v}) < 0$.*

Proof. Put $p_1 = \det \text{br}_{\tilde{g}_1}(\tilde{v})$, $p_2 = \det \text{br}_{\tilde{q}}(\tilde{v})$, $p_3 = \det \text{br}_{\tilde{q}}(\tilde{g}_1)$. By Lemma 2.1 and [4], (3), we get $p_1 = -2$. Using formula (2) in [4] for the determinant of $(\tilde{g}_1 \tilde{q})$ regarded as a subgraph of $\tilde{L}$, we get $p_2 = -2p_3 + 1 < 0$ since $p_3 > 0$.

Lemmas 2.5 and 2.6 are proved analogously to Lemmas 2.3 and 2.4.

Lemma 2.5. *Suppose that $\tilde{q} \in \langle \tilde{g}_2 \tilde{v} \rangle$ is nodal, $\det(\tilde{q}\tilde{g}_2) = 2$ and $\det \delta(\tilde{q}\tilde{g}_2) = 1$. Then $\tilde{q}$ is inessential for $\tilde{g}_2$.*

Lemma 2.6. *Suppose that $\tilde{q}$ satisfies the hypotheses of Lemma 2.5, and that $\det \text{br}_{\tilde{s}}(\tilde{v}) = 1$ for any vertex $\tilde{s} \in \langle \tilde{q}\tilde{v} \rangle$ which is essential for $\tilde{g}_1$. (This condition holds automatically if $(\tilde{v}\tilde{q})$ is linear.) Then $\det \text{br}_{\tilde{q}}(\tilde{v}) < 0$.*

Lemma 2.7. *Let $\tilde{q}_1 \in \langle \tilde{v}\tilde{g}_2 \rangle$ be essential for $\tilde{g}_2$, and $\det \delta(\tilde{q}_1 \tilde{g}_2) = \det(\tilde{q}_1 \tilde{g}_2) = 1$. Let $\tilde{q}_2 \in \langle \tilde{v}\tilde{q}_1 \rangle$ be nodal, and $\det \delta(\tilde{q}_2 \tilde{q}_1) = 1$, $\det(\tilde{q}_2 \tilde{g}_2) = d(\tilde{q}_1) + 1$. Then $\tilde{q}_2$ is inessential for $\tilde{g}_2$.*

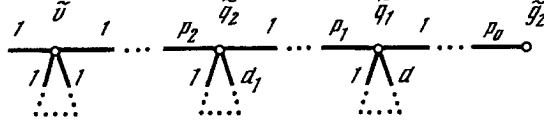


Figure 3

Proof. Assume the opposite. Put $d = d(\tilde{q}_1)$, $d_1 = d(\tilde{q}_2)$, $p_0 := \det \text{br}_{\tilde{g}_2}(\tilde{v})$, $p_1 := \det \text{br}_{\tilde{q}_1}(\tilde{v})$, $p_2 := \det \text{br}_{\tilde{q}_2}(\tilde{v})$. Since $d \neq 1$, we have $\det \text{br}_{\tilde{q}_1}(\tilde{g}_2) = 1$. Since $d_1 \neq 1$, we have $\det \text{br}_{\tilde{q}_2}(\tilde{g}_2) = 1$. Hence the splice diagram of L is as in Fig. 3. Applying [4], (2) to $(\tilde{q}_1\tilde{g}_2)$ and $(\tilde{q}_2\tilde{g}_2)$ regarded as subgraphs of $\tilde{L}$, we get

$$-1 = p_0 - dp_1, \quad -d - 1 = p_0 - d^2 d_1 p_2 \quad (2.3)$$

It follows from [4], (3) that

$$1 = k_{\tilde{g}_2} = -2 - p_0 + (d - 1)(p_1 - 1) + d(d_1 - 1)(p_2 - 1) + dd_1 \Sigma, \quad (2.4)$$

where Σ is the remainder in [4], (3). Substituting (2.3) in (2.4), we get $p_2 = -d_1 + d_1 \Sigma$. This contradicts [4], Lemma 2.

Lemma 2.8. *Let $\tilde{q}_1$ be the unique nodal vertex of $\langle \tilde{v}\tilde{g}_2 \rangle$, and let $\det(\tilde{q}_1\tilde{g}_2) = 1$. Then $\det \text{br}_{\tilde{q}_1}(\tilde{v}) < -1$.*

Proof. If $\tilde{q}_1$ is inessential for $\tilde{g}_2$, then $\det \text{br}_{\tilde{g}_2}(\tilde{v}) = -3$ by [4], (3). Hence $\det \text{br}_{\tilde{q}_1}(\tilde{v}) = -3 \det \text{br}_{\tilde{q}_1}(\tilde{g}_2) + 1 < -1$. If $\tilde{q}_1$ is essential for $\tilde{g}_2$, then $\det \text{br}_{\tilde{q}_1}(\tilde{g}_2) = 1$ by [4], Lemma 3. Applying [4], (2) to $(\tilde{q}_1\tilde{g}_2)$ regarded as a subgraph of $\tilde{L}$ and [4], (3) to $k_{\tilde{g}_2}$, we get $\det \text{br}_{\tilde{q}_1}(\tilde{v}) = -d(\tilde{q}_1) - 1 < -1$.

In the next three lemmas we suppose that $\tilde{v} \in \langle \tilde{g}_1\tilde{g}_2 \rangle$. Let $\tilde{k}_1, \dots, \tilde{k}_n$ be all essential vertices of $\langle \tilde{g}_1\tilde{g}_2 \rangle$. Put $d_j = d(\tilde{k}_j)$, $r_j := \det \text{br}_{\tilde{k}_j}(\tilde{g}_2)$, $l_j := \det \text{br}_{\tilde{k}_j}(\tilde{g}_1)$, $r_{\tilde{g}_1} := \det \text{br}_{\tilde{g}_1}(\tilde{g}_2)$, $l_{\tilde{g}_2} := \det \text{br}_{\tilde{g}_2}(\tilde{g}_1)$. Then the splice diagram of $\tilde{L}$ is as in Fig. 4.

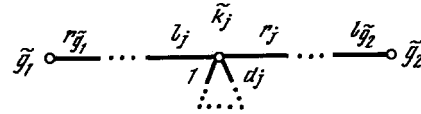


Figure 4

Lemma 2.9. *Let $\tilde{k}_1, \dots, \tilde{k}_n$ be enumerated in the order in which they appear as we move from $\tilde{g}_1$ to $\tilde{g}_2$, and let $\tilde{v} \in \langle \tilde{k}_n\tilde{g}_2 \rangle$. Put $\mu_i = \det(\tilde{k}_i\tilde{g}_2)$. If $\mu_i = 2$ for $i > p$, then*

$$0 = 3 + \det \tilde{L}_\infty + 2d_1 \dots d_p - \sum_{j=1}^p \mu_j (d_j - 1) \prod_{q < j} d_q. \quad (2.5)$$

(In particular, if $\mu_j = 2$ for $j \geq 1$, then $\det \tilde{L}_\infty = -5$.) If, moreover, $\mu_p = 1$ and $\mu_i = d_p + 1$ for $s < i < p$, then

$$0 = 3 + \det \tilde{L}_\infty + d_1 \dots d_s(d_p + 1) - \sum_{j=1}^s \mu_j(d_j - 1) \prod_{q < j} d_q. \quad (2.6)$$

(In particular, if $\mu_j = d_p + 1$ for $1 \leq j < p$, then $\det \tilde{L}_\infty = -4 - d_p$.)

Proof. Using [4], (3) for $k_{\tilde{g}_2}$, we get $l_{\tilde{g}_2} = -3$. Since $\min(d_1, \dots, d_n) > 1$, it follows from [4], Lemma 3 that $l_1 = \dots = l_n = 1$. Applying [4], (2) to $(\tilde{k}_i \tilde{g}_2)$ regarded as a subgraph of $\tilde{L}$, we get

$$r_i = -\frac{1}{3}(d_i \prod_{q > i} d_q^2 - \mu_i), \quad i = 1 \dots n-1, \quad r_n = -\frac{1}{3}(d_n - \mu_n). \quad (2.7)$$

Applying [4], (2) to $\tilde{L}_\infty$ regarded as a subgraph of $\tilde{L}$, we get

$$r_{\tilde{g}_1} = \frac{1}{3}(\det \tilde{L}_\infty - d_1^2 \dots d_n^2). \quad (2.8)$$

Since

$$\sum_{j=1}^n (d_j - 1) \prod_{k < j} d_k = d_1 \dots d_n - 1, \quad (2.9)$$

the formula [4], (3) for $k_{\tilde{g}_1}$ takes the form

$$0 = -1 - d_1 \dots d_n - r_{\tilde{g}_1} + \sum_{j=1}^n (d_j - 1) r_j \prod_{k < j} d_k.$$

Combining this with (2.7), (2.8), we get

$$0 = -1 + \frac{1}{3}(\prod_{q=1}^n d_q^2 - \det \tilde{L}_\infty) - \prod_{q=1}^n d_q - \frac{1}{3} \sum_{j=1}^n (d_j - 1)(d_j \prod_{q > j} d_q^2 - \mu_j) \prod_{k < j} d_k.$$

Using (2.9) again, we get

$$0 = -3 - \det \tilde{L}_\infty - 2d_1 \dots d_n + \sum_{j=1}^n \mu_j(d_j - 1) \prod_{q < j} d_q. \quad (2.10)$$

Putting $\mu_i = 2$ for $i > p$ and using (2.9), we obtain (2.5). Moreover, if $\mu_{s+1} = \dots = \mu_{p-1} = d_p + 1$ and $\mu_p = 1$, then (2.5) and (2.9) yield (2.6).

The next two lemmas are proved analogously to Lemma 2.9.

Lemma 2.10. Let $\tilde{k}_1, \dots, \tilde{k}_n$ be enumerated in the order in which they appear as we move from $\tilde{g}_2$ to $\tilde{g}_1$, and let $\tilde{v} \in (\tilde{k}_n \tilde{g}_1)$. Put $\mu_i = \det(\tilde{k}_i \tilde{g}_1)$. If $\mu_i = 1$ for $i > p$, then

$$0 = 4 + \det \tilde{L}_\infty + d_1 \dots d_p - \sum_{j=1}^p \mu_j(d_j - 1) \prod_{q < j} d_q. \quad (2.11)$$

(In particular, if $\mu_j = 1$ for $j \geq 1$, then $\det \tilde{L}_\infty = -5$.)

Lemma 2.11. *Let $\tilde{k}_1, \dots, \tilde{k}_n$ be enumerated in the order in which they appear as we move from $\tilde{g}_1$ to $\tilde{g}_2$, and let $\tilde{v} \in (\tilde{k}_{n-1}\tilde{k}_n)$ and $\det(\tilde{k}_n\tilde{g}_2) = 1$. Put $\mu_i = \det(\tilde{k}_i\tilde{g}_2)$. If $\mu_j = d_n + 1$ for $p < j \leq n - 1$, then*

$$0 = 1 + \det \tilde{L}_\infty + (d_1 \dots d_p + d_n)(d_n + 1) - \sum_{j=1}^p \mu_j(d_j - 1) \prod_{q < j} d_q. \quad (2.12)$$

(In particular, if $\mu_j = d_n + 1$ for $1 \leq j \leq n - 1$, then $\det \tilde{L}_\infty = -1 - (d_n + 1)^2$.)

The last lemma of this section is concerned with the position of the root $\tilde{v}$ in $\tilde{L}$.

Lemma 2.12. *Suppose that $\tilde{a} \in \langle \tilde{g}_1\tilde{g}_2 \rangle$. Put $Q_1 := \text{br}_{\tilde{a}}(\tilde{g}_1)$, $Q_2 := \text{br}_{\tilde{a}}(\tilde{g}_2)$. Let Q_3 be the subgraph that consists of all connected components of $\delta(\tilde{g}_1\tilde{g}_2)$ incident to $\tilde{a}$ (see Fig. 5). Suppose that $\det \tilde{L}_\infty \leq 0$, $\det Q_3 > 0$ and*

$$\det(\tilde{a}\tilde{g}_i) \leq (\det \delta(\tilde{a}\tilde{g}_i))^2 \det Q_3 \quad (2.13)$$

for some $i = 1, 2$. Then $\tilde{v} \in Q_1 \cup Q_2$.

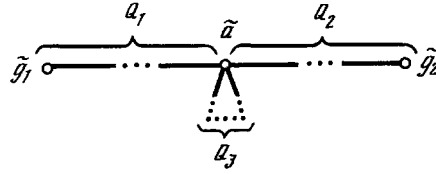


Figure 5

Proof. If $\tilde{v} \notin Q_1 \cup Q_2$, then $\det Q_i > 0$, $\det(\tilde{a}\tilde{g}_i) > 0$, $\det \delta(\tilde{a}\tilde{g}_i) > 0$, $i = 1, 2$. Suppose that (2.13) holds for $i = 1$. Put $\Delta_i = \det \text{br}_{\tilde{g}_i}(\tilde{a})$, $i = 1, 2$. Applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{a})$, we get

$$\Delta_1 \det(\tilde{a}\tilde{g}_2) = \det Q_2 \det \tilde{L}_\infty - \det(\tilde{a}\tilde{g}_1) \det Q_3 (\det \delta(\tilde{a}\tilde{g}_2))^2.$$

Since the right-hand side is negative and $\det(\tilde{a}\tilde{g}_2) > 0$, we have $\Delta_1 < 0$. But, using [4], (2) for $(\tilde{a}\tilde{g}_1)$ regarded as a subgraph of $\tilde{L}$, we see that

$$-\det(\tilde{a}\tilde{g}_1) = \det Q_1 \Delta_1 - \det Q_2 \det Q_3 (\det \delta(\tilde{a}\tilde{g}_1))^2.$$

Combining this with (2.13) and the inequality $\det Q_2 \geq 1$, we get $\det Q_1 \Delta_1 \geq 0$, so $\Delta_1 \geq 0$. This contradiction completes the proof.

§ 3. The structure of the graph $\tilde{L}$

We regard $\tilde{L}_\infty$ as consisting of forks connected by “edge type” subgraphs. (The definition is given below.) The purpose of this section is to demonstrate that there are only finitely many possibilities for $\tilde{L}_\infty$ and to describe them all. We recall that vertices of $\tilde{L}_\infty$ are marked by a tilde, and their images under F are denoted by the same letters but without tilde. If $a \in L$, then we denote by a' the set of all $\tilde{a} \in F^{-1}(a)$ such that $F|_{\tilde{a}}$ is non-constant.

Since the topological degree of F equals four, we get

$$\sum_{\tilde{a} \in a'} \text{Deg}(\tilde{a}) = 4. \quad (3.1)$$

Since $\tilde{L}$ is a tree, we have the following lemma.

Lemma 3.1. *Suppose that a, b, c are vertices of L , and $b \in (ac)$. Suppose also that $\tilde{a} \in a'$, $\tilde{b} \in b'$, and $(\tilde{a}\tilde{b})$ is a subgraph of constant degree (see [4], Definition 4). Then*

$$\sum_{\tilde{c} \in c' \cap \text{br}_{\tilde{a}}(\tilde{b})} \text{Deg}(\tilde{c}) \geq \text{Deg}(\tilde{a}\tilde{b}). \quad (3.2)$$

Definition 3.2. A connected subgraph $\tilde{Q} \subset \tilde{L}_\infty$ is called an “edge type” subgraph if the following conditions hold.

- a) At most two vertices of $\tilde{L}_\infty + \tilde{g}_2$ are incident to $\tilde{Q}$ and F is non-constant on these vertices.
- b) If $\tilde{Q}$ is incident to only one such vertex $\tilde{a} \in \tilde{L}_\infty + \tilde{g}_2$, then $\det \tilde{Q} = \frac{n(\tilde{a})}{\text{Deg } \tilde{Q}} \det F(\tilde{Q})$. If $\tilde{Q}$ is incident to two such vertices $\tilde{a}, \tilde{b} \in \tilde{L}_\infty + \tilde{g}_2$, then $\det \tilde{Q} = \frac{n(\tilde{a})n(\tilde{b})}{\text{Deg } \tilde{Q}} \det F(\tilde{Q})$.
- c) If $\tilde{a} \in \tilde{L}_\infty + \tilde{g}_2$ is incident to $\tilde{Q}$, and $\tilde{T} \subset \tilde{Q}$ is the linear chain incident to $\tilde{a}$, then $\text{Deg } \tilde{T} = \text{Deg } \tilde{Q}$.

Convention on figures. In the figures below, we indicate the pair $(m(\tilde{s}), n(\tilde{s}))$ near each vertex $\tilde{s}$. To the left (right) of $\tilde{s}$ are situated those subgraphs incident to $\tilde{s}$ whose images under F lie to the left (right) of s . If $\tilde{s} \notin h'$, then below $\tilde{s}$ are situated those subgraphs incident to $\tilde{s}$ whose images under F lie below s (see Fig. 1).

Unless otherwise stated, the edges lying above (below) a vertex $\tilde{s} \in h'$ correspond to the “edge type” subgraphs incident to $\tilde{s}$ whose images under F lie above (below) s , and the numbers on the edges are their degrees.

We recall the definitions of a fork and a maximal subgraph of constant degree. (They are the same as in [4].)

Definition 3.3. a) A vertex $\tilde{l} \subset \tilde{L}_\infty$ is called a *fork* if $m(\tilde{l}) > 1$. b) A subgraph of constant degree is said to be *maximal* if it is not contained in another subgraph of constant degree.

The following obvious proposition shows that vertices of $\tilde{L}_\infty$ incident to a maximal subgraph of constant degree are forks.

Proposition 3.4. *Let $\tilde{s} \in \tilde{L}_\infty$ be a nodal vertex. Then the following conditions are equivalent.*

- a) $\nu(\tilde{s}) = \nu(s)$. (Here $\nu(s)$ is the valency of s in $\Gamma_{L, g_1 + g_2}$.)
- b) $m(\tilde{s}) = 1$.
- c) $\text{Deg}(\tilde{C}) = \text{Deg}(\tilde{s}) = n(\tilde{s})$ for any linear chain $\tilde{C}$ incident to $\tilde{s}$.

Let $\tilde{Q}$ be a maximal subgraph of constant degree. By Proposition 3.4, the splice diagrams of $\tilde{Q}$ and $F(\tilde{Q})$ differ only in their weights. Hence conditions 3.2 a), c) hold if $\tilde{Q} \cap h' = \emptyset$ (see Fig. 1). Moreover, if $\tilde{Q}$ is not incident to $\tilde{g}_1$, then condition 3.2 b) follows from [4], Lemma 5. If $\tilde{Q}$ is incident to $\tilde{g}_1$, then we have $n(\tilde{g}_1) = 1$ by Lemma 1.5, and condition 3.2 b) also follows from [4], Lemma 5. This proves the following assertion.

Assertion 3.5. *Let $\tilde{Q}$ be a maximal subgraph of constant degree with $\tilde{Q} \cap h' = \emptyset$. Then $\tilde{Q}$ is an “edge type” subgraph.*

Let us prove an auxiliary statement.

Assertion 3.6. *Let $\tilde{Q}$ be an “edge type” subgraph incident to only one vertex $\tilde{a}$ of the graph $\tilde{L}_\infty + \tilde{g}_2$. Put $Q = F(\tilde{Q})$ and $\tilde{x} = \tilde{a} \cap \tilde{Q}$. If $\det Q = 1$, then $\det \tilde{Q} = m_{\tilde{x}}(\tilde{a}) = 1$, $n(\tilde{a}) = \text{Deg } \tilde{Q}$. (The number $m_{\tilde{x}}$ is defined in [4], Definition 3.)*

Proof. By condition 3.2 b), $\det \tilde{Q} = \frac{n(\tilde{a})}{\text{Deg } \tilde{Q}}$. By condition 3.2 c) and [4], Proposition 1, we have $\text{Deg } \tilde{Q} = \text{Deg } \tilde{T} = m_{\tilde{x}}(\tilde{a})n(\tilde{a})$ for any linear chain $\tilde{T} \subset \tilde{Q}$ incident to $\tilde{a}$. We get $\det \tilde{Q} = m_{\tilde{x}}(\tilde{a}) = 1$, $\text{Deg } \tilde{Q} = n(\tilde{a})$.

Lemma 3.7. a) *Let $c \in U \cup h$ be a nodal vertex. Then every connected component $\tilde{\Lambda} \subset F^{-1}(U_c)$ is an “edge type” subgraph incident to one vertex $\tilde{c} \in c'$ of the graph $\tilde{L}_\infty + \tilde{g}_2$. (We recall that $U_h = U$.)*

b) *If $c \in U$, then $\det \tilde{\Lambda} = 1$ and all but two of the branches of $\tilde{L}_\infty$ incident to $\tilde{c}$ belong to $F^{-1}(U_c)$. One of these two branches contains h' and the other is linear and has positive determinant.*

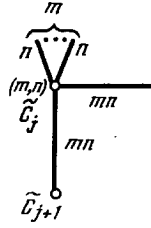


Figure 6

Proof. Let $c_1, \dots, c_p = h$ be all nodal vertices of $U + h$ enumerated from the top downwards (see Fig. 1).

a) We argue by induction. 1) Suppose that $\tilde{\Lambda} \subset F^{-1}(U_{c_1})$. Then $\tilde{\Lambda}$ is linear and incident to either one vertex $\tilde{c}_1 \in c'$ or two vertices $\tilde{c}_1 \in c'$ and $\tilde{g}_1$. Conditions 3.2 a), c) hold automatically, and condition 3.2 b) follows from [4], Lemma 5 and Lemma 1.5.

2) Suppose that a) holds for $c_1, \dots, c_j$. We claim that a) holds for c_{j+1} . Let $\tilde{c}_j$ be a vertex in c'_j , let $\tilde{\Lambda} \subset F^{-1}(U_{c_{j+1}})$ be the connected component that contains $\tilde{c}_j$, and put $m := m(\tilde{c}_j)$, $n := n(\tilde{c}_j)$. By Proposition 1.3, $\det U_{c_j} = 1$. Using the induction hypothesis, Assertion 3.6 and formulae (4) and (5) of [4], we see that $\tilde{\Lambda}$ corresponds to Fig. 6, $\tilde{\Lambda}$ is incident to a unique vertex $\tilde{c}_{j+1} \in c'_{j+1}$ of the graph $\tilde{L}_\infty + \tilde{g}_2$ (hence conditions 3.2 a), c) hold), and $\tilde{c}_j$ is incident to exactly m connected components of $F^{-1}(U_{c_j})$. The determinant of each component equals one. Put $\tilde{R} := \tilde{\Lambda} \cap F^{-1}(R_{c_j})$. The subgraphs $\tilde{R}$ and $(\tilde{c}_j \tilde{c}_{j+1})$ are linear since R_{c_j} and $(c_j c_{j+1})$ are linear. By Proposition 1.3 and [4], Lemma 5, we have $\det \tilde{R} = \frac{\det R_{c_j}}{m} > 0$,

$\det(\tilde{c}_{j+1}\tilde{c}_j) = \frac{n(\tilde{c}_{j+1})}{m} \det(c_{j+1}c_j)$. Applying [4], (6) to $\tilde{c}_j$ regarded as a vertex of $\tilde{\Lambda}$, we get

$$\det \tilde{\Lambda} = \frac{m}{n} \cdot \frac{1}{m} \cdot \frac{n(\tilde{c}_{j+1})}{m} \det U_{c_{j+1}} = \frac{n(\tilde{c}_{j+1})}{nm} \det U_{c_{j+1}}.$$

Since $\text{Deg } \tilde{\Lambda} = mn$, we see that condition 3.2 b) holds.

Assertion b) is proved similarly to step 2) in a).

We put $U' = F^{-1}(U)$.

Corollary 3.8. a) *Every connected component $\tilde{U} \subset U'$ is an “edge type” subgraph incident to one vertex of the graph $\tilde{L}_\infty + \tilde{g}_2$.*

b) *If $\tilde{U}$ is incident to $\tilde{g}_1$, then $\tilde{v} \notin \delta(\tilde{g}_1\tilde{h})$, and all nodal vertices of $\delta(\tilde{g}_1\tilde{v}) \cap \tilde{U}$ are inessential for $\tilde{g}_1$.*

Proof. Assertion a) follows from Lemma 3.7 a). Assertion b) follows from Lemmas 2.3, 2.4 and 3.7 b).

We now study the structure of $\tilde{L}$ near h' . Suppose that $\tilde{h} \in h'$ and either $m(\tilde{h}) = 1$ or L is of type 3. We define a subgraph $\Delta_{\tilde{h}}$ as follows. If $m(\tilde{h}) \neq 1$, then $\Delta_{\tilde{h}}$ consists of $\tilde{h}$ together with all maximal subgraphs of constant degree incident to $\tilde{h}$ and all connected components of U' incident to $\tilde{h}$. If $m(\tilde{h}) = 1$, then $\Delta_{\tilde{h}}$ consists of the maximal subgraph of constant degree that contains $\tilde{h}$ and all connected components of U' incident to $\tilde{h}$.

Assertion 3.9. a) *Suppose that $\tilde{h} \in h'$ is a fork and L is of type 1 or 3. Then $\tilde{h}$ is incident to exactly $m(\tilde{h})$ connected components of U' . The determinant of each component equals one. Suppose that $\tilde{h} \cap U' = \{\tilde{p}_1, \dots, \tilde{p}_n\}$. Then $m_{\tilde{p}_1}(\tilde{h}) = \dots = m_{\tilde{p}_n}(\tilde{h}) = 1$.*

b) $\Delta_{\tilde{h}}$ is an “edge type” subgraph.

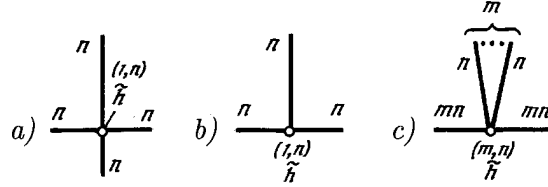


Figure 7

Proof. a) This follows from Assertion 3.6, Corollary 3.8 a) and [4], (4) because $\det U = 1$.

b) If $\tilde{h}$ is not a fork, then Proposition 3.4 implies that $\Delta_{\tilde{h}}$ is as in Fig. 7, a), b). If $\tilde{h}$ is a fork, then a) and formulae (4), (5) of [4] imply that $\Delta_{\tilde{h}}$ is as in Fig. 7, c). Clearly, vertices of $\tilde{L}_\infty + \tilde{g}_2$ incident to $\Delta_{\tilde{h}}$ are either to the left or to the right of $\tilde{h}$, so conditions 3.2 a), c) hold. Condition 3.2 b) is verified in the same way as in part 2) of the proof of Lemma 3.7.

Thus we have shown that the following subgraphs are of “edge type”:

- a) Maximal subgraphs of constant degree that do not intersect h' .
- b) Connected components of U' .
- c) The subgraphs $\Delta_{\tilde{h}}$ defined before Assertion 3.9.

In what follows, $\tilde{L}_\infty$ will be “constructed” from forks and subgraphs of the form a), b), c). The components described in Assertion 3.9 a) contribute nothing to the right-hand side of [4], (5), and their determinants are equal to one. Therefore they will not appear in the remaining figures in §3.

Definition 3.10. Suppose that $\tilde{a} \in \tilde{L}_\infty$ does not belong to a subgraph of the form a), b), c). (Then $\tilde{a}$ is a fork, $\tilde{a} \notin U'$, and L is of type 1 or 2 if $a = h$.) We denote by $U_{\tilde{a}}$ the subgraph of $\tilde{L}_\infty$ that contains $\tilde{a}$ and all the subgraphs a), b), c) incident to $\tilde{a}$.

In the rest of §3, we depict the subgraphs $U_{\tilde{a}}$ as consisting of the vertex $\tilde{a}$ and right, left and lower edges incident to $\tilde{a}$. The lower edges of these subgraphs correspond to connected components of $F^{-1}(D_a)$. We note that $D_h = U$ if L is of type 2 (see the convention in §1). We also do not label connected components of U' if L is of type 1.

The following simple assertion is useful in the study of $U_{\tilde{a}}$.

Assertion 3.11. a) Let $\tilde{Q}$ be a left branch of $U_{\tilde{a}}$. Then $\det F(\tilde{Q})$ and $\det D_a$ are coprime, $\det L_a > 1$, $\det D_a > 1$, $\det R_a = 1$.

b) All lower branches of $U_{\tilde{a}}$ and those left branches of $U_{\tilde{a}}$ that are also branches of $\tilde{L}_\infty$ have pairwise coprime determinants except possibly for one pair, of which one of the branches is incident to $\tilde{g}_1$.

Proof. a) This follows from Proposition 1.3, 2) and Corollary 1.4.

b) This follows from [4], Lemma 2.

Lemma 3.12. 1) $U_{\tilde{a}}$ corresponds to one of the diagrams a), b), c), f), g), j)-t) in Fig. 8.

2) In the cases b), s) in Fig. 8, the left branch of $U_{\tilde{a}}$ is incident to $\tilde{b}$, and $n(\tilde{b}) = m(\tilde{b}) = 2$.

3) In the cases a), l), r) in Fig. 8, we have $\tilde{a} \in h'$ and one of the lower branches of $U_{\tilde{a}}$ is incident to $\tilde{g}_1$. In the case l) in Fig. 8 we have $\text{Deg}(\tilde{a}\tilde{g}_1) = 1$.

4) In the case c) in Fig. 8, either $\tilde{a}$ is to the right of $\tilde{h}$, or one of the left branches of $U_{\tilde{a}}$ is incident to some fork.

5) In the case k) in Fig. 8, one of the left branches of degree 1 in $U_{\tilde{a}}$ is incident to $\tilde{g}_1$.

6) If $U_{\tilde{a}}$ corresponds to one of the diagrams b), s), t) in Fig. 8, then $\tilde{a}$ is to the right of $\tilde{h}$.

Proof. We omit the details since they are analogous to [4], Lemma 8. Using Definition 3.2, we express the determinants of the branches of $U_{\tilde{a}}$ in terms of their images under F . Combining formulae (4), (5) of [4] with Assertion 3.11, a) and Assertion 3.9, we see that $U_{\tilde{a}}$ corresponds to one of the diagrams in Fig. 8 and, in the cases b), s), the left branch of $U_{\tilde{a}}$ is incident to a fork $\tilde{b}$ such that $n(\tilde{b}) = m(\tilde{b}) = 2$. This proves 2).

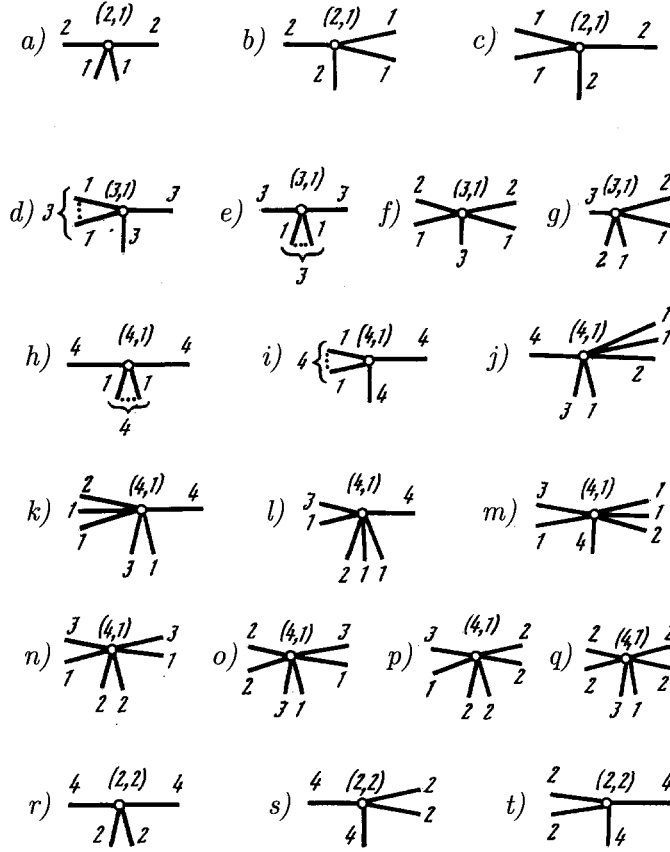


Figure 8

By Definition 3.2, b), the determinants of the lower branches in the subgraphs a), e), h), r) in Fig. 8 and the lower branches of degree 1 in the subgraph l) are equal to $\det D_a$. By Assertion 3.11, the cases e), h) are impossible and 3) is valid.

According to Definition 3.2 b), the determinants of the left branches of degree 1 in $U_{\tilde{a}}$ that are also branches of $\tilde{L}_{\infty}$ are equal to $\det L_a$. It follows from (3.2) that if $\text{Deg } \tilde{a} = 4$, then the left branches of degree 1 in $U_{\tilde{a}}$ are also branches of $\tilde{L}_{\infty}$. Combining these facts with Assertion 3.11, we see that i) in Fig. 8 is impossible, (4), (5) are valid, and in the case d) one of the left branches of $U_{\tilde{a}}$ is incident to some fork $\tilde{c}$.

Consider d) in Fig. 8. By (3.1) and (3.2), we get $\text{Deg } \tilde{c} = 2$, so $U_{\tilde{c}}$ corresponds to b). But the left branch of $U_{\tilde{c}}$ is incident to the fork $\tilde{b}$ of degree 4, so all branches of $\tilde{L}$ that begin to the left of $\tilde{a}$ contain $\tilde{b}$. Since $\tilde{L}$ is a tree, we obtain a contradiction. Thus d) is impossible.

We have shown that the cases d), e), h), i) are impossible. This proves 1).

It remains to prove 6). Assume the opposite: either a is to the left of h or $\tilde{a} \in h'$. We must show that the diagrams b), s), t) in Fig. 8 are impossible in these cases.

Suppose that $U_{\tilde{a}}$ corresponds to t). As in 4), we see that one of the left branches is incident to some fork $\tilde{b}$. Since $\text{Deg } \tilde{a} = 4$, it follows from (3.1) and (3.2) that $U_{\tilde{b}}$ corresponds to one of the diagrams a) or c). Since a is not to the right of h , this contradicts 3) and 4).

Assume that $U_{\tilde{a}}$ corresponds to s). We see from 1) that the left branch of $U_{\tilde{a}}$ is incident to the fork $\tilde{b}$ with $m(\tilde{b}) = n(\tilde{b}) = 2$, so $U_{\tilde{b}}$ corresponds to r) or t). Since a is not to the right of h , this contradicts previous results.

Suppose that $U_{\tilde{a}}$ corresponds to b). According to 1), the left branch of $U_{\tilde{a}}$ is incident to the fork $\tilde{b}$ with $m(\tilde{b}) = n(\tilde{b}) = 2$, so $U_{\tilde{b}}$ corresponds to s). This contradicts previous results. Thus 6) is established. This completes the proof of the lemma.

Lemma 3.13. *The diagrams b), p), q), s) in Fig. 8 are impossible. If $U_{\tilde{a}}$ corresponds to c), then both left branches of $U_{\tilde{a}}$ are also branches of $\tilde{L}_\infty$. One of these branches is incident to $\tilde{g}_1$.*

Proof. Assume that p) or q) corresponds to some $U_{\tilde{a}}$. Since $\text{Deg } \tilde{a} = 4$, it follows that the branches of $\tilde{L}_\infty$ beginning to the right of $\tilde{a}$ lie entirely to the right of $\tilde{a}$. Let $\tilde{R}$ be the right branch of $\tilde{L}_\infty$ such that $\tilde{R}$ is incident to $\tilde{a}$ and non-incident to $\tilde{g}_2$. By Lemma 3.12, either $\tilde{R}$ is a branch of $U_{\tilde{a}}$, or $\tilde{R} = U_{\tilde{b}}$ and $U_{\tilde{b}}$ corresponds to a). If $\tilde{R}$ is a branch of $U_{\tilde{a}}$, then $\det \tilde{R} = \frac{\det R_a}{2} = \frac{1}{2}$ by Assertion 3.11 a) and Definition 3.2 b). This is impossible. Suppose that $\tilde{R} = U_{\tilde{b}}$. We denote the right branch of $U_{\tilde{b}}$ by $\tilde{R}'$. As above, we get $\det \tilde{R}' = \frac{\det R_b}{2} = \frac{1}{2}$, which is impossible. Thus p), q) are impossible.

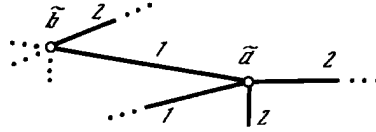


Figure 9

Suppose that c) corresponds to some $U_{\tilde{a}}$, and one of its left branches is incident to a fork $\tilde{b} \neq \tilde{a}$. If $\text{Deg } \tilde{b} = 2$, then $U_{\tilde{b}}$ corresponds to b). This contradicts the fact that $\tilde{L}_\infty$ is a tree. Assume that $\text{Deg } \tilde{b} = 3$. Then $U_{\tilde{b}}$ corresponds to f) or g). Pasting $U_{\tilde{a}}$ and $U_{\tilde{b}}$ along the “edge” $(\tilde{a}\tilde{b})$, we obtain Fig. 9. Let $R_{\tilde{a}}$ be the branch of $\tilde{L}_\infty$ that begins to the right of $\tilde{a}$, and let $R_{\tilde{b}}$ be the branch of $\tilde{L}_\infty$ that begins to the right of $\tilde{b}$ and does not contain $\tilde{a}$. We see from (3.1) and (3.2) that $R_{\tilde{a}}$ ($R_{\tilde{b}}$) lies on one side of $\tilde{a}$ ($\tilde{b}$). As in the study of p), q), we obtain that the determinant of the branch of $R_{\tilde{a}}$, $R_{\tilde{b}}$ that is not incident to $\tilde{g}_2$ equals $\frac{1}{2}$, which is impossible. Thus both branches of $U_{\tilde{a}}$ are also branches of $\tilde{L}_\infty$. By Lemma 3.12, this is possible only if one of them is incident to $\tilde{g}_1$.

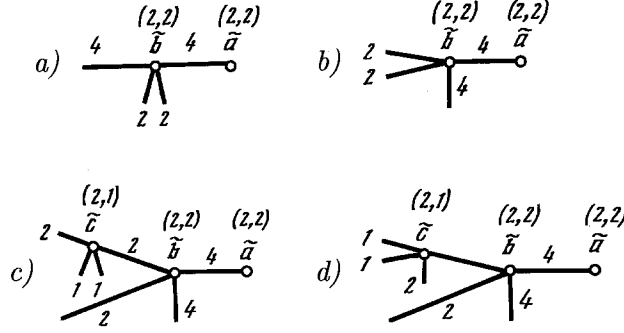


Figure 10

Suppose that $U_{\tilde{a}}$ corresponds to s). Then the left branch of $U_{\tilde{a}}$ is incident to a fork $\tilde{b}$ such that $n(\tilde{b}) = m(\tilde{b}) = 2$. Let $L_{\tilde{a}}$ be the left branch of $\tilde{L}_{\infty}$ beginning at $\tilde{a}$. It follows from Lemma 3.12 that the branch $L_{\tilde{a}}$ corresponds to one of a)-d) in Fig. 10.

We claim that $\det L_a$ is even. Indeed, consider the case a) in Fig. 10. (The other cases are treated in the same way.) We denote the determinant of the left (resp. right, lower) branch of $U_{\tilde{b}}$ by d_1 (resp. d_2, d_3). (The determinants of the lower branches of $U_{\tilde{b}}$ are equal by Definition 3.2, b).) By Definition 3.2, $\det L_b = 2d_1$, $\det(ba) = d_2$, $\det D_b = d_3$. Corollary 1.4 implies that d_3 is odd. Applying [4], (6) to $\tilde{b}$, we get $\det L_a = \frac{2}{d_3} \det L_{\tilde{a}}$. Since d_3 is odd, $\det L_a$ is even.

Let d_7 be the determinant of the left branch of $U_{\tilde{a}}$. By Definition 3.2, b), $\det D_a = 2d_7$, so $\det D_a$ is also even, which contradicts Proposition 1.3, 2) and the fact that $\det L_a$ is odd. Thus s) in Fig. 8 is impossible.

Consider b) in Fig. 8. The left branch of the corresponding graph is incident to a fork $\tilde{b}$ such that $m(\tilde{b}) = n(\tilde{b}) = 2$. By Fig. 8, $U_{\tilde{b}}$ must correspond to s) in Fig. 8, which is impossible. Thus b) in Fig. 8 is impossible.

We associate with the graph $\tilde{L}_{\infty} + \tilde{g}_2$ the following diagram: replace the subgraphs $U_{\tilde{a}}$ corresponding to a), c) in Fig. 8 by edges of degree 2 and the other subgraphs $U_{\tilde{a}}$ by the corresponding diagrams in Fig. 8.

Lemma 3.14. $\tilde{L}_{\infty} + \tilde{g}_2$ corresponds to one of the diagrams in Fig. 11.

Proof. By Lemma 1.5, the degree of the linear chain incident to $\tilde{g}_2$ is 2. Combining this with Lemmas 3.12, 3.13, we get the result.

Lemma 3.15. Suppose that $a \in L - U$ is nodal, $\tilde{a} \in a'$, Δ is the branch of $\tilde{L}_{\infty}$ of degree 2 incident to $\tilde{a}$ such that $F(\Delta) = R_a$, and Δ is incident to $\tilde{g}_2$ and non-incident to $\tilde{g}_1$. Then $\tilde{v} \notin \delta(\tilde{a}\tilde{g}_2)$ and every nodal vertex of $\langle \tilde{v}\tilde{g}_2 \rangle \cap \Delta$ is inessential.

Proof. If Δ contains no forks, then Proposition 3.4 says that the splice diagrams of Δ and $F(\Delta)$ differ only in their weights. If Δ contains forks, then Lemmas 3.12 and 3.13 yield that Δ is a subgraph of the graph $\Delta_{\tilde{h}}$ defined before Assertion 3.9, the only possible fork of Δ is $\tilde{h} \in h'$ and, in this case, the determinants of the connected

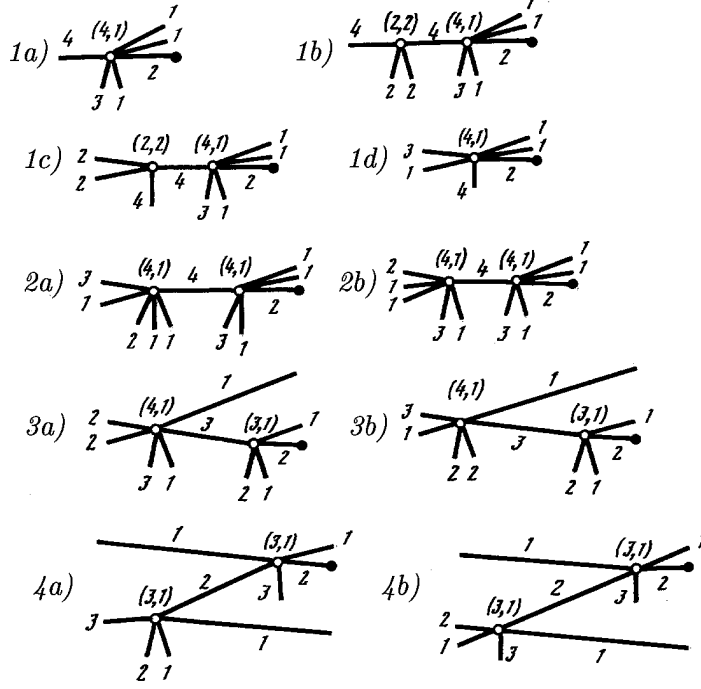


Figure 11

components of U' incident to $\tilde{h}$ are equal to one. Using (6) and Lemma 5 of [4], Proposition 1.3 and Definition 3.2, we obtain that $\det \delta(\tilde{a}\tilde{g}_2) > 0$, all connected components of $\delta(\tilde{a}\tilde{g}_2) - U'$ are linear, and every nodal vertex $\tilde{s} \in \langle \tilde{a}\tilde{g}_2 \rangle$ that is not a fork satisfies $\det(\tilde{s}\tilde{g}_2) = 2$. Applying Lemmas 3.7, b), 2.5 and 2.6, we get the result.

Lemma 3.16. *Let $L_{\tilde{a}}$ be a left branch of $U_{\tilde{a}}$ such that $L_{\tilde{a}}$ is not incident to $\tilde{g}_1$, $\tilde{v} \notin L_{\tilde{a}}$, $F(L_{\tilde{a}}) = L_a$. If $h \notin L_a$, then $L_{\tilde{a}}$ is linear. If $h \in L_a$, then L_h and (ha) are linear, L is of type 3 and $\det L_h > 0$.*

Proof. If $h \notin L_a$, then $L_{\tilde{a}}$ is a maximal subgraph of constant degree and, by Proposition 3.4, the splice diagrams of $L_{\tilde{a}}$ and L_a can differ only in their weights. If $h \in L_a$, then $L_{\tilde{a}}$ is one of the subgraphs $\Delta_{\tilde{h}}$ defined before Assertion 3.9, and $L_{\tilde{a}}$ corresponds to one of the diagrams in Fig. 7. Thus the only possible fork of $L_{\tilde{a}} - U'$ is $\tilde{h}$, and in this case L is of type 3. Put $L_{\tilde{h}} = F^{-1}(L_h) \cap L_{\tilde{a}}$. Since $\tilde{v} \notin L_{\tilde{h}}$, we have $\det L_{\tilde{h}} > 0$, and $\det L_h > 0$ by [4], Lemma 5. It remains to show that if $\tilde{b}$ is a nodal vertex of $L_{\tilde{a}} - U'$ such that $m(\tilde{b}) = 1$, then $b = h$ and L is of type 3. Put $L_{\tilde{b}} := F^{-1}(L_b) \cap L_{\tilde{a}}$, $D_{\tilde{b}} := F^{-1}(D_b) \cap L_{\tilde{a}}$. (Recall that $\det L_{\tilde{h}} > 0$ if L is of type 2.) Combining formula (6) and Lemma 5 of [4] with Definition 3.2, we get $\det L_{\tilde{b}} = \det L_b$ and $\det D_{\tilde{b}} = \det D_b$. Since $\tilde{v} \notin L_{\tilde{a}}$, we have $\min(\det L_b, \det D_b) = 1$. By Proposition 1.3, 2), this is possible only if $b = h$ and L is of type 3.

§ 4. Prohibition of diagrams 2a), 2b) in Fig. 11

We denote the right (resp. left) vertex of the diagram by $\tilde{a}$ (resp. $\tilde{b}$).

Lemma 4.1. *The cases 2a), 2b) are impossible.*

Proof. We work out the case 2a) in detail. The case 2b) is treated analogously. By Lemma 3.12, 3), $\tilde{g}_1$ is incident to one of the lower branches of degree 1 of $U_{\tilde{a}}$. Hence all edges of the diagram correspond to “edge type” subgraphs. Let l_a^i be the determinant of the left branch of degree i in $U_{\tilde{a}}$, let d_a^i (resp. d_b^i) be the determinant of the lower branch of degree i in $U_{\tilde{a}}$ (resp. in $U_{\tilde{b}}$), and let r_b^i be the determinant of the right branch of degree i in $U_{\tilde{b}}$. By Definition 3.2,

$$\begin{aligned} l_a^1 &= 3l_a^3 = \det L_a, & d_a^1 &= 2d_a^2 = \det D_a, \\ d_b^1 &= 3d_b^3 = \det D_b, & r_b^1 &= r_b^2 = \det R_b = 1. \end{aligned} \quad (4.1)$$

Applying (4.1) and [4], Lemma 3 to $\tilde{a}$, $\tilde{b}$, we get

$$\tilde{v} \notin \text{br}_{\tilde{a}}(\tilde{b}), \quad \det \text{br}_{\tilde{a}}(\tilde{b}) = 1, \quad d_b^3 = 1, \quad d_b^1 = \det D_b = 3, \quad \det \text{br}_{\tilde{b}}(\tilde{g}_2) = 1. \quad (4.2)$$

Applying [4], (6) to $\tilde{b}$ regarded as a vertex of $(\tilde{a}\tilde{g}_2)$, we get $\det(\tilde{a}\tilde{g}_2) = 1$. (Analogous calculations were used in [4], Lemma 15.) Using [4], (2) for $(\tilde{b}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{a}}(\tilde{b})$, we get $1 = 1 - 3 \det(\tilde{a}\tilde{b})(\det \delta(\tilde{b}\tilde{g}_2))^2$. This contradicts the fact that $\tilde{v} \notin \text{br}_{\tilde{a}}(\tilde{b})$.

§ 5. Prohibition of diagrams 1a)-1d) in Fig. 11

Let $\tilde{h} \in h'$ be a vertex such that $(\tilde{h}\tilde{g}_1)$ is a connected component of U' .

Prohibition of 1a). Let $\tilde{a}$ be the vertex of the diagram. By Assertion 3.9 and Definition 3.10, $\tilde{L}_\infty$ corresponds to one of the diagrams in Fig. 12.

Put $L_{\tilde{a}} = F^{-1}(L_a)$ and let D_a^i be the connected component of degree i in $F^{-1}(D_a)$ incident to $\tilde{a}$. By Definition 3.2,

$$\det L_a = 4 \det L_{\tilde{a}}, \quad \det D_a^1 = \det D_a = 3 \det D_a^3. \quad (5.1)$$

By Proposition 1.3, 2), $\det R_a = 1$. Hence Definition 3.2 implies that

$$\det(F^{-1}(R_a) - (\tilde{a}\tilde{g}_2)) = 1 \quad (5.2)$$

If $(\tilde{a}\tilde{g}_2)$ is an “edge type” subgraph, then Definition 3.2 yields that $\det(\tilde{a}\tilde{g}_2)_\infty = 1$. (We recall that $n(\tilde{g}_2) = 2$.) If $(\tilde{a}\tilde{g}_2)$ is not an “edge type” subgraph, then it corresponds to 10) in Fig. 12 and $\det(\tilde{a}\tilde{g}_2)_\infty = \det U$ by [4], (6). Hence

$$\det(\tilde{a}\tilde{g}_2)_\infty = \begin{cases} \det U & \text{if } \tilde{L} \text{ corresponds to 10) in Fig. 12,} \\ 1 & \text{otherwise.} \end{cases} \quad (5.3)$$

Applying [4], (6) to $\tilde{a}$ and using (5.1)–(5.3) and Assertion 3.9, we get the following assertion.

Assertion 5.1. 1) *If $\tilde{L}$ corresponds to one of the diagrams 1)–5), 8), 9), 11), 12), then $\det D_a = 3$, $\det \tilde{L}_\infty = -1$.*

2) *If $\tilde{L}$ corresponds to one of the diagrams 6), 7), then $\det \tilde{L}_\infty = -\frac{1}{3} \det U$.*

3) *If $\tilde{L}$ corresponds to 10), then $\det \tilde{L}_\infty = -\det U$, $\det D_a = 3$.*

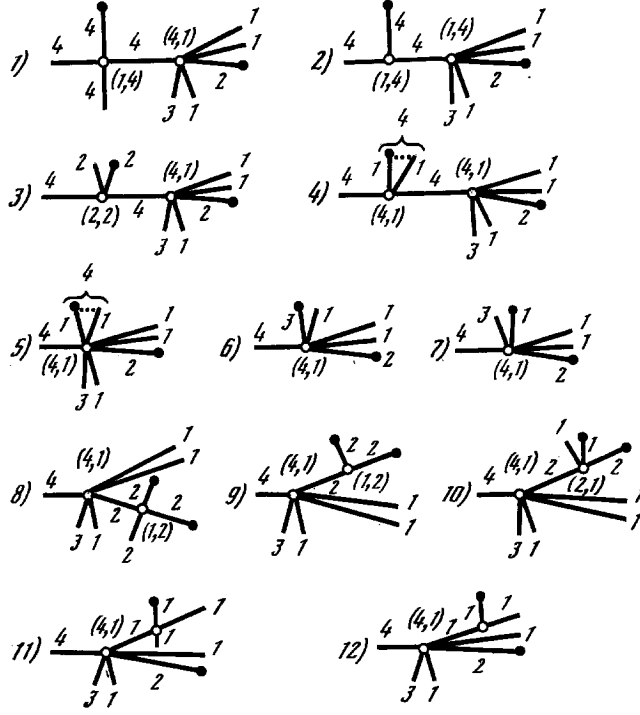


Figure 12

Assertion 5.2. 1) Suppose that $\tilde{q}$ is a nodal vertex of $\langle \tilde{a}\tilde{g}_2 \rangle$, $m(\tilde{q}) = 1$ and $\tilde{g}_1 \notin \langle \tilde{g}_2\tilde{q} \rangle$. Then $\det(\tilde{q}\tilde{g}_2) = 2$.

2) Suppose that $\tilde{q}$ is a nodal vertex of $L_{\tilde{a}}$ and $q \in (ha)$. Then $\det(\tilde{q}\tilde{g}_2) = 4$.

3) Suppose that $\tilde{q}$ satisfies 1) or 2), and let $D_{\tilde{q}}$ be a connected component of $F^{-1}(D_q)$ incident to $\tilde{q}$. Then $\det D_{\tilde{q}} > 1$.

Proof. 1) Note that $\tilde{q}$ is not a fork, $n(\tilde{q}) = n(\tilde{g}_2) = 2$, $(\tilde{q}\tilde{g}_2)$ is an “edge type” subgraph. Combining this with [4], Lemma 5, we get the result.

2) Since $\tilde{q}$ is not a fork, it follows that $n(\tilde{q}) = 4$ and $(\tilde{q}\tilde{a})$ is a subgraph of constant degree. By [4], (8), $\det(\tilde{a}\tilde{q}) = \det(aq)$. Then we apply [4], (6) to $\tilde{a}$ regarded as a vertex of $(\tilde{q}\tilde{g}_2)$ and combine this with (5.1)–(5.3) and Assertion 5.1.

3) By Definition 3.2 and Assertion 1.3, $\det D_{\tilde{q}} = \det D_q > 1$.

Lemma 5.3. The cases 1)–4) in Fig. 12 are impossible.

Proof. Assume the opposite. Put $L_{\tilde{h}} = F^{-1}(L_h)$. (The vertex $\tilde{h}$ is defined at the beginning of § 5.) The scheme of the proof is as follows: we first show that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{a}) \cup (\tilde{a}\tilde{g}_2)$. Then we show that $\tilde{v}$ cannot belong to any of these sets.

Combining Definition 3.2, (5.1)–(5.3), Assertion 5.1 and [4], (6) for $\tilde{a}$ regarded as a vertex of $(\tilde{h}\tilde{g}_2)$, we obtain

$$\det L_{\tilde{h}} = \frac{n(\tilde{h})}{4} \det L_h, \quad \det(\tilde{a}\tilde{h}) = \frac{n(\tilde{h})}{4} \det(ah), \quad \det(\tilde{h}\tilde{g}_2) = n(\tilde{h}) \det R_h. \quad (5.4)$$

1) We claim that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{a}) \cup (\tilde{a}\tilde{g}_2)$. Let us first prove that $\det L_{\tilde{h}} > 0$. By (5.4), it suffices to check that $\det L_h > 0$. If L is of type 1 or 2, this follows from Proposition 1.3, 2). Suppose that L is of type 3. Using Proposition 1.3, 2) and Assertion 5.1, we have $\det R_a = 1$, $\det L_a > 1$, $\det D_a = 3$. Using [4], Lemma 4, we have $\det R_h > 0$, $\det(ha) > 0$, $\det \delta(ha) > 0$. Applying [4], (2) to (ha) regarded as a subgraph of L , we get

$$\det L_h = \frac{1}{3(\det \delta(ha))^2} (\det L_a \det R_h + \det(ha)) > 0$$

Since $\det L_h > 0$, we see from (5.1)–(5.3), Assertion 5.1 and (5.4) that the determinants of all branches of $\tilde{L}_\infty$ incident to $\tilde{a}$ or $\tilde{h}$ are positive and $\det \tilde{L}_\infty < 0$, $\det(\tilde{a}\tilde{g}_2) = 1$. Applying Lemma 2.12 to $\tilde{a}$, we get $\tilde{v} \in (\tilde{a}\tilde{g}_2) \cup L_{\tilde{a}}$. If $\det U = 1$, then $\det(\tilde{h}\tilde{g}_1) = 1$. If $\det U \neq 1$, then $\tilde{L}$ corresponds to 2) in Fig. 12, $\det(\tilde{h}\tilde{g}_2) = 4$ and $\det \delta(\tilde{h}\tilde{g}_2) \geq 3$. Applying Lemma 2.12 to $\tilde{h}$, we get $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. Thus, $\tilde{v} \in ((\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)) \cap ((\tilde{a}\tilde{g}_2) \cup L_{\tilde{a}}) = (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{a}) \cup (\tilde{a}\tilde{g}_2)$.

2) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. By Lemma 3.15 and Corollary 3.8, $(\tilde{a}\tilde{g}_2)$ is linear, $\det \delta(\tilde{v}\tilde{g}_1) = 1$. By [4], (3) and Lemma 2.1, $\det \text{br}_{\tilde{g}_1}(\tilde{a}) = -2$. By [4], Lemma 3, $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$. Applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{a})$ and combining this with (5.2), (5.3) and Assertion 5.1, we get $3 \det L_a = 1$, which is impossible.

3) Assume that $\tilde{v} \in (\tilde{h}\tilde{a})$. Using Lemma 2.7, Lemma 2.8, Corollary 3.8, Lemma 3.15, (5.3) and Assertion 5.2, we see that $(\tilde{a}\tilde{g}_2)$ and $(\tilde{v}\tilde{a})$ are linear, $\det \delta(\tilde{h}\tilde{g}_1) = 1$.

If $\det U = 1$, then $\det(\tilde{h}\tilde{g}_1) = 1$ and $\tilde{h}$ is inessential by Lemma 2.3. If $\det U \neq 1$ then $\tilde{L}$ corresponds to 2) in Fig. 12 and $\det(\tilde{h}\tilde{g}_2) = 4$ by Proposition 1.3, 2) and (5.4). In the notation of Lemma 2.11, we see that $\tilde{k}_n = \tilde{a}$, $\mu_1 = \dots = \mu_{n-1} = 4$, $d_n = 3$, $\det \tilde{L}_\infty = -1$ independently of the value of $\det U$. This contradicts (2.12).

4) Assume that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. By Lemma 3.15, $(\tilde{v}\tilde{g}_2)$ is linear. By Corollary 3.8, $\det \delta(\tilde{h}\tilde{g}_1) = 1$. By Assertions 5.1 and 5.2, we have (in the notation of Lemma 2.9) $\det \tilde{L}_\infty = -1$, $\mu_1 = \dots = \mu_{p-1} = 4$, $\mu_p = \det(\tilde{a}\tilde{g}_2) = 1$, $d_p = 3$, $\mu_{p+1} = \dots = \mu_n = 2$. Here the number p is chosen to satisfy $\tilde{k}_p = \tilde{a}$. This contradicts (2.6).

Lemma 5.4. *The cases 5), 6), 7) in Fig. 12 are impossible.*

Proof. Assume the opposite. The scheme of the proof is similar to that of Lemma 5.3. We first show that $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$. Then we show that $\tilde{v}$ cannot belong to either of these sets.

1) It follows from (5.1)–(5.4) and Assertion 5.1 that the determinants of all branches of $\tilde{L}_\infty$ incident to $\tilde{a}$ are positive, $\det \tilde{L}_\infty < 0$ and $\det(\tilde{a}\tilde{g}_2) = 1$. Applying Lemma 2.12 to $\tilde{a}$, we get $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$.

2) Assume that $\tilde{v} \in (\tilde{a}\tilde{g}_1)$. The case 5) in Fig. 12 is analogous to 2) in the proof of Lemma 5.3. If $\tilde{L}$ corresponds to one of the diagrams 6), 7), then we get the following facts as in step 2) of the proof of Lemma 5.3: $(\tilde{a}\tilde{g}_2)$ is linear, $\det(\tilde{a}\tilde{g}_2) = 1$, $\det \text{br}_{\tilde{g}_1}(\tilde{a}) = -2$.

If $\tilde{a}$ is essential, then Lemma 3 of [4] implies that $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$. Applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{a})$ and combining this with Assertion 5.1,

we get $-2 = -\frac{1}{3} \det U - \frac{1}{3} (\det U)^2 \det L_{\tilde{a}}$, which contradicts $\det L_{\tilde{a}} > 0$. If $\tilde{a}$ is inessential, then $\tilde{L}$ corresponds to 7) in Fig. 12 and $\det U = 3$. In the notation of Lemma 2.10, we have $\mu_1 = \dots = \mu_n = 1$, $\det \tilde{L}_\infty = -1$ by Corollary 3.8 and Assertion 5.1. This contradicts (2.11).

3) Assume that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. Then Corollary 3.8 implies that $\det \delta(\tilde{a}\tilde{g}_1) = 1$. By Lemma 3.15 and Assertion 5.2, the subgraph $(\tilde{v}\tilde{g}_2)$ is linear.

If $\tilde{L}$ corresponds to 5) in Fig. 12, then $\det(\tilde{a}\tilde{g}_1) = 1$ and, by Lemma 2.3, $\det D_a^1 = \det D_a^3 = 1$, which contradicts Assertion 5.1.

Suppose that $\tilde{L}$ corresponds to one of the diagrams 6) or 7) in Fig. 12. Then $\det \tilde{L}_\infty = -\frac{1}{3} \det U$ and $\det(\tilde{a}\tilde{g}_2) = 1$. If $\tilde{a}$ is inessential, then $\tilde{L}$ corresponds to 7) and $\det \tilde{L}_\infty = -1$. By Assertion 5.2, we have $\mu_1 = \dots = \mu_n = 2$ in the notation of Lemma 2.9. This contradicts (2.5). If $\tilde{a}$ is essential, then (in the notation of Lemma 2.9) $\mu_1 = \det(\tilde{a}\tilde{g}_2) = 1$, $\mu_2 = \dots = \mu_n = 2$, and $d_1 = d(\tilde{a})$ is equal to one of the numbers $\det U$, $\frac{1}{3} \det U$, $\det L_{\tilde{a}}$. Moreover, $d_1 = \det L_{\tilde{a}}$ only if $\tilde{L}$ corresponds to 7) and $\det U = 3$. Substituting these data in (2.6), we see that $d_1 = \frac{1}{3} \det U - 4$. This contradicts any of the three possibilities for d_1 .

Lemma 5.5. *The case 8) in Fig. 12 is impossible.*

Proof. Assume the opposite. Put $D_{\tilde{h}} := F^{-1}(D_h) \cap (\tilde{a}\tilde{g}_2)$. Proposition 1.3, 2), Definition 3.2 and Assertion 5.1 imply that $\det \tilde{L}_\infty = -1$, $D_{\tilde{h}}$ is linear, $\det D_{\tilde{h}} = \det D_h > 1$, $\det(\tilde{h}\tilde{g}_1) = 1$, $\det(\tilde{h}\tilde{g}_2) = 2$. If $\tilde{v} \notin (\tilde{h}\tilde{g}_2)$, then $\tilde{v} \in D_{\tilde{h}}$ by Lemmas 2.5 and 3.15. This contradicts Lemma 2.6. If $\tilde{v} \in (\tilde{h}\tilde{g}_2)$, then Lemma 2.3 and Corollary 3.8 yield a contradiction to the inequality $\det D_{\tilde{h}} > 1$.

Lemma 5.6. *The case 9) is impossible. The case 10) in Fig. 12 is impossible if $\det U = 1$.*

Proof. Assume the opposite. The scheme of the proof is as in Lemma 5.3. By Assertion 5.1, $\det \tilde{L}_\infty = -1$. According to Definition 3.2 and Proposition 1.3, 2), in the case 9) we have either $\det(\tilde{h}\tilde{g}_1) = 1$, or $\det(\tilde{h}\tilde{g}_2) = 2$. In the case 10), if $\det U = 1$, then we have $\det(U' \cap (\tilde{a}\tilde{g}_2)) = 1$.

1) We claim that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. It suffices to prove that $\tilde{v} \notin \tilde{h} \cup \text{br}_{\tilde{h}}(\tilde{a})$. Indeed, in the case 9) we have $\tilde{L}_\infty = (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2) \cup (\tilde{h} \cup \text{br}_{\tilde{h}}(\tilde{a}))$, and in the case 10) we have $\det(\tilde{L}_\infty - \tilde{h}) > 0$ and $\det(\tilde{h}\tilde{g}_1) = 1$ provided that $\tilde{v} \notin \tilde{h} \cup \text{br}_{\tilde{h}}(\tilde{a})$. Then Lemma 2.12 implies that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$.

Assume that $\tilde{v} \in \{\text{br}_{\tilde{h}}(\tilde{a}) \cup \tilde{h}\}$. By Corollary 3.8, Lemma 3.15 and Assertion 5.2 we see that $(\tilde{h}\tilde{g}_2)$ is linear, $\det \delta(\tilde{h}\tilde{g}_1) = 1$. Using Proposition 1.3, 2) and Lemmas 2.3 and 2.5, we see that the following holds.

$$\begin{aligned} \text{If } \det U = 1, \text{ then } \det(\tilde{h}\tilde{g}_1) &= 1, & \det \text{br}_{\tilde{h}}(\tilde{g}_2) &= 1. \\ \text{If } \det U \neq 1, \text{ then } \det(\tilde{h}\tilde{g}_2) &= 2, & \det \text{br}_{\tilde{h}}(\tilde{g}_1) &= 1, \end{aligned} \quad (5.5)$$

and in this case $\tilde{L}$ corresponds to 9) in Fig. 12.

a) We claim that $\tilde{v} \in (\tilde{a}\tilde{h})$. By (5.3), $\det(\tilde{a}\tilde{g}_2)_\infty = 1$. If $\det U = 1$, then (5.2), (5.3), (5.5) and [4], (2) for $(\tilde{h}\tilde{g}_2)$ regarded as a subgraph of $(\tilde{a}\tilde{g}_1)$ yield that

$\det(\tilde{h}\tilde{g}_2)\det(\tilde{a}\tilde{g}_1) = 1 - \det(\tilde{h}\tilde{a})$. Since $\det R_h > 0$ and $\det(ha) > 0$, we have $\det(\tilde{h}\tilde{g}_2) > 0$, $\det(\tilde{h}\tilde{a}) > 0$ by [4], Lemma 5. Hence $\det(\tilde{a}\tilde{g}_1) \leq 0$. Analogously, $\det(\tilde{a}\tilde{g}_2) \leq 0$ if $\det U \neq 1$. Hence $\tilde{v} \in (\tilde{a}\tilde{h})$.

b) We claim that $(\tilde{v}\tilde{h})$ is linear. This is obvious if $(\tilde{a}\tilde{h})$ is linear. Suppose that there is a nodal vertex $\tilde{s} \in (\tilde{a}\tilde{h})$ such that $(\tilde{s}\tilde{h})$ is linear. Put $D_{\tilde{s}} = F^{-1}(D_s) \cap (\tilde{g}_2\tilde{a})$. Then $\det(\tilde{s}\tilde{g}_2)_\infty = 2$, $\det D_{\tilde{s}} = \det D_s > 1$. Using (5.5) and formula (2) of [4], we get $\det(\tilde{s}\tilde{g}_1)\det(\tilde{h}\tilde{g}_2) = 2 - \det(\tilde{s}\tilde{h})$ if $\det(\tilde{h}\tilde{g}_1) = 1$. If $\det(\tilde{h}\tilde{g}_2) = 2$, then $\det(\tilde{s}\tilde{g}_2)\det(\tilde{h}\tilde{g}_1) = 2 - 2\det(\tilde{h}\tilde{s})$. Combining this with Lemma 4 of [4], we see that either $\tilde{v} \in (\tilde{s}\tilde{g}_1)_\infty$, or $\det(\tilde{s}\tilde{g}_1) = \det(\tilde{h}\tilde{g}_1) = 1$. In the second case we use Lemmas 2.3, 2.4 and the inequality $\det D_{\tilde{s}} > 1$ to conclude that $\tilde{v} \in (\tilde{s}\tilde{g}_1)_\infty$. By the choice of $\tilde{s}$, we see that $\tilde{v} \in (\tilde{s}\tilde{h})$, and $(\tilde{v}\tilde{h})$ is linear.

c) We now claim that $\tilde{v} \notin \tilde{h} \cup \text{br}_{\tilde{h}}(\tilde{a})$. If $\det U = 1$, then $\det \text{br}_{\tilde{g}_1}(\tilde{h}) = -2$ by (5.5) and [4], (3). Using [4], (2) for $(\tilde{h}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{h})$, we get $\det \text{br}_{\tilde{h}}(\tilde{a}) = 2\det(\tilde{h}\tilde{g}_2) - 1 \geq 1$, and Lemma 2.12 implies that $\tilde{v} \notin \{\text{br}_{\tilde{h}}(\tilde{a}) \cup \tilde{h}\}$. If $\det U \neq 1$, then $\det \text{br}_{\tilde{g}_2}(\tilde{h}) = -3$ by (5.5) and [4], (3). We have $\det(\tilde{h}\tilde{g}_1) = \det U > 1$, $\det(\tilde{h}\tilde{g}_2) = 2$. Combining this with formula (2) of [4] for $(\tilde{h}\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{g}_2}(\tilde{h})$, we get $\det \text{br}_{\tilde{h}}(\tilde{a}) = \frac{1}{2}(3\det U - 1) > 2$, and Lemma 2.12 implies that $\tilde{v} \notin \{\text{br}_{\tilde{h}}(\tilde{a}) \cup \tilde{h}\}$. This proves 1).

2) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. Then $(\tilde{h}\tilde{g}_2)$ is linear. If $\det U = 1$, then $\det(\tilde{h}\tilde{g}_1) = 1$. If $\det U \neq 1$, then $\det(\tilde{h}\tilde{g}_2) = 2$ and $\tilde{h}$ is inessential by Lemma 2.5. Combining this with Corollary 3.8, we get (in the notation of Lemma 2.10) $\mu_1 = \dots = \mu_n = 1$ and $\det \tilde{L}_\infty = -1$. This contradicts (2.11).

3) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_2)$. By Lemma 3.15, $(\tilde{v}\tilde{g}_2)$ is linear. Reasoning as above and using Lemma 2.3 and Assertions 5.1, 5.2, we get (in the notation of Lemma 2.9) $\mu_1 = \dots = \mu_n = 2$ and $\det \tilde{L}_\infty = -1$. This contradicts (2.5).

Lemma 5.7. *The case 10) in Fig. 12 is impossible if $\det U \neq 1$.*

Proof. Assume the opposite. The scheme of the proof is as in Lemma 5.3. Put $d := \det U$. Let $\tilde{U}$ be a connected component of U' incident to $\tilde{h}$ and non-incident to $\tilde{g}_1$. Using Lemma 5 of [4], (5.3) and Assertion 5.1, we get

$$\det \tilde{U} = \det(\tilde{h}\tilde{g}_1) = \det(\tilde{a}\tilde{g}_2) = d > 1, \quad \det \tilde{L}_\infty = -d, \quad \det(\tilde{h}\tilde{g}_2) = 1. \quad (5.6)$$

1) We claim that $\tilde{v} \in (\tilde{h}\tilde{g}_2) \cup (\tilde{h}\tilde{g}_1)$. It suffices to show that $\tilde{v} \notin \text{br}_{\tilde{h}}(\tilde{a})$. For then $\det \text{br}_{\tilde{h}}(\tilde{a}) > 0$ and we see from (5.6) and Lemma 2.12 that $\tilde{v} \in (\tilde{h}\tilde{g}_2) \cup (\tilde{h}\tilde{g}_1)$.

Suppose that $\tilde{v} \in \text{br}_{\tilde{h}}(\tilde{a})$. Applying [4], Lemma 3 to $\tilde{h}$, we get $\det \text{br}_{\tilde{h}}(\tilde{g}_1) = \det \text{br}_{\tilde{h}}(\tilde{g}_2) = 1$. By Corollary 3.8 and Lemma 3.15, we see that $(\tilde{h}\tilde{g}_2)$ is linear and $\det \delta(\tilde{g}_1\tilde{h}) = 1$. Using [4], (2) for the subgraphs $(\tilde{g}_1\tilde{h})$ and $(\tilde{g}_2\tilde{h})$ of the graph $\tilde{L}$, we get

$$-1 = \det \text{br}_{\tilde{g}_2}(\tilde{h}) - d \det \text{br}_{\tilde{h}}(\tilde{a}), \quad -d = \det \text{br}_{\tilde{g}_1}(\tilde{h}) - d \det \text{br}_{\tilde{h}}(\tilde{a}), \quad (5.7)$$

By Lemma 2.1 and [4], (3),

$$k_{\tilde{g}_1} = 0 = -2 - \det \text{br}_{\tilde{g}_1}(\tilde{h}) + \Sigma, \quad k_{\tilde{g}_2} = 1 = -2 - \det \text{br}_{\tilde{g}_2}(\tilde{h}) + \Sigma.$$

Here Σ is the remainder in formula (3) of [4]. Hence $\det \text{br}_{\tilde{g}_2}(\tilde{h}) - \det \text{br}_{\tilde{g}_1}(\tilde{h}) = -1$. Substituting this in (5.7), we get $d = 0$, which is impossible.

2) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_2)$. In the notation of Lemma 2.9, we have $\det \tilde{L}_\infty = -d$, $\mu_1 = 1$, $d_1 = d$, $\mu_2 = \dots = \mu_n = 2$. This contradicts (2.6).

3) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. In the notation of Lemma 2.10, we have $\det \tilde{L}_\infty = -d$, $\mu_1 = d$, $d_1 = d$, $\mu_2 = \dots = \mu_n = 1$. Substituting this in (2.11), we get $d(d-1) = 4$. But this equation has no integer solutions.

Lemma 5.8. *The cases 11), 12) in Fig. 12 are impossible.*

Proof. Assume the opposite. Put $R_{\tilde{h}} = \text{br}_{\tilde{a}}(\tilde{h}) \cap F^{-1}(R_h)$.

1) Suppose that $\tilde{v} \in (\tilde{a}\tilde{g}_1)$. By Lemma 3.15, $\det \delta(\tilde{a}\tilde{g}_2) = 1$. Hence L is of type 3, and the subgraphs $(\tilde{a}\tilde{h})$ and $R_{\tilde{h}}$ are linear. Since $\det U = 1$, we have $\det(\tilde{h}\tilde{g}_1) = 1$. Reasoning as in Corollary 3.8, we get $\tilde{v} \in \langle \tilde{a}\tilde{g}_1 \rangle$ and $\det \delta(\tilde{v}\tilde{g}_1) = 1$. In the notation of Lemma 2.10, we have $\mu_1 = \dots = \mu_n = 1$. This contradicts (2.11) since $\det \tilde{L}_\infty = -1$.

2) Suppose that $\tilde{v} \notin (\tilde{a}\tilde{g}_1)$. By Proposition 1.3, 2) and Corollary 3.8, every essential nodal vertex $\tilde{s} \in \langle \tilde{a}\tilde{g}_1 \rangle$ satisfies $\det(\tilde{s}\tilde{g}_1) = 1$. Hence Lemma 2.3 implies that $\tilde{v} \in D_a^1$. By Lemma 2.7, this contradicts the equation $\det D_a^1 = 3$ and the linearity of D_a^1 .

Using Lemmas 5.3–5.8, we obtain the following lemma.

Lemma 5.9. *The case 1a) in Fig. 11 is impossible.*

Prohibition of 1b)-1d) in Fig. 11.

Lemma 5.10. *The cases 1b), 1c) are impossible.*

Proof. Assume the opposite. Let $\tilde{a}$ (resp. $\tilde{b}$) be the right (resp. left) vertex of the diagram, and let $D_a^i, i = 1, 3$ be the connected component of $F^{-1}(D_a)$ of degree i incident to $\tilde{a}$. Put $d = \det(\tilde{b}\tilde{g}_1)$. As in (5.1) and (5.4), it is easy to prove that

$$\begin{aligned} \det D_a^3 &= 1, & \det D_a^1 &= \det D_a = 3, & \det(\tilde{a}\tilde{g}_2) &= 1, \\ \det(\tilde{b}\tilde{g}_2) &= 2, & \det(\tilde{a}\tilde{b}) &= \frac{1}{2} \det(ab) > 0, & & (5.8) \\ \det \text{br}_{\tilde{a}}(\tilde{b})_\infty &> 0, & \det \tilde{L}_\infty &< 0. \end{aligned}$$

As in (5.4), every nodal vertex $\tilde{s} \in (\tilde{b}\tilde{a})$ satisfies $\det(\tilde{s}\tilde{g}_2) = 4$.

1) We claim that $\tilde{v} \in (\tilde{b}\tilde{a}) \cup (\tilde{a}\tilde{g}_2)$. By (5.8) and Lemma 2.12, $\tilde{v} \in (\tilde{a}\tilde{g}_2) \cup (\tilde{a}\tilde{g}_1)$. It suffices to prove that $\tilde{v} \notin (\tilde{a}\tilde{g}_1) - (\tilde{a}\tilde{b})$.

Assume the opposite. By [4], Lemma 3, $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$. Using (5.8) and formula (2) of [4] for $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{b}}(\tilde{g}_2)$, we get $\det \text{br}_{\tilde{b}}(\tilde{g}_2) = 2 - 3 \det(\tilde{a}\tilde{b})(\det \delta(\tilde{a}\tilde{g}_2))^2 < 0$. This contradicts our assumption.

2) We claim that $\det \delta(\tilde{b}\tilde{g}_1) = 1$. By Corollary 3.8, this is true if $\tilde{L}$ corresponds to 1b) in Fig. 11. Suppose that $\tilde{L}$ corresponds to 1c) in Fig. 11. Applying Lemma 3.16 to the branch of degree 2 that begins at $\tilde{b}$ and is non-incident to $\tilde{g}_1$, we see that L is of type 3, and the subgraphs $(b\tilde{h})$ and L_b are linear. Thus $(\tilde{b}\tilde{h})$ is linear,

and $\det(\tilde{h}\tilde{g}_1) = 1$ by Assertion 3.6. Since $\tilde{v} \notin (\tilde{b}\tilde{g}_1)$, we see by Corollary 3.8 and Lemma 2.3 that $\det\delta(\tilde{b}\tilde{g}_1) = 1$. Since L is of type 3, the branches of degree 2 of $\tilde{L}_\infty$ incident to $\tilde{b}$ are “edge type” subgraphs. Using (5.8) and [4], (6), we get $\det\tilde{L}_\infty = -d$.

3) Assume that $\tilde{v} \in (\tilde{b}\tilde{a})$. It follows from Lemmas 2.7, 2.8 and 3.15 that $(\tilde{v}\tilde{a})$ and $(\tilde{a}\tilde{g}_2)$ are linear. Since $\det\delta(\tilde{b}\tilde{g}_1) = 1$, (5.8) and Assertion 5.2 imply that (in the notation of Lemma 2.11) $\mu_1 = \det(\tilde{b}\tilde{g}_2) = 2$, $d_1 = d$, $\mu_2 = \dots = \mu_{n-1} = 4$, $d_n = 3$, $\det\tilde{L}_\infty = -d$. Substituting these data in (2.12), we get $0 = 15 + d$. This is impossible since $d > 0$.

4) Assume that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. By Lemma 3.15, $(\tilde{v}\tilde{g}_2)$ is linear. In the notation of Lemma 2.9 we get $\mu_1 = 2$, $d_1 = d$, $\det\tilde{L}_\infty = -d$, $\mu_2 = \dots = \mu_{p-1} = 4$, $\mu_p = 1$, $d_p = 3$, $\mu_{p+1} = \dots = \mu_n = 2$ for some positive integer p . Substituting these data in (2.6), we get $0 = 5 + d$. This is impossible since $d > 0$.

Lemma 5.11. *The case 1d) in Fig. 11 is impossible.*

Proof. Assume the opposite. We denote the vertex of the diagram by $\tilde{a}$. Then $\tilde{L}_\infty = U_{\tilde{a}}$ (see Lemma 3.12). As in the prohibition of 5), 8)-12) in Fig. 12, we prove that $\tilde{g}_1$ is non-incident to any right branch of $U_{\tilde{a}}$, and if $a = h$, then L is of type 2 or 3. Hence $\tilde{g}_1$ is incident to either the lower branch or one of the left branches of $U_{\tilde{a}}$. Put $D_{\tilde{a}} = F^{-1}(D_a)$. Let $L_{\tilde{a}}^i$ be the connected component of degree i in $F^{-1}(L_a)$. As in (5.1), (5.3) and Assertion 5.1, we get

$$\begin{aligned} \det(\tilde{a}\tilde{g}_2) &= 1, & \det D_a &= 4 \det \tilde{D}_a, & \det L_a &= \det L_a^1 = 3 \det L_a^3, \\ \det \tilde{L}_\infty &= \begin{cases} -1 & \text{if } \tilde{g}_1 \text{ is incident to } D_{\tilde{a}}, \\ -\frac{1}{3} \det L_a & \text{if } \tilde{g}_1 \text{ is incident to } L_a^1 \text{ or } L_a^3. \end{cases} \end{aligned} \quad (5.9)$$

Applying Lemma 2.12 to $\tilde{a}$, we get $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$.

1) Assume that $\tilde{g}_1$ is incident to one of the subgraphs L_a^1 , L_a^3 . We see from Lemma 3.16 that L_h and (ha) are linear, L is of type 3, $\det L_h > 0$. Put $L_{\tilde{h}} = F^{-1}(L_h) \cap (\tilde{a}\tilde{g}_1)$. We have $\det(\tilde{h}\tilde{g}_1) = 1$, $L_{\tilde{h}}$ and $(\tilde{h}\tilde{a})$ are linear, $\det L_{\tilde{h}} > 0$. Combining Corollary 3.8 and Lemmas 2.3, 2.4, we see that $\tilde{v} \notin \delta(\tilde{g}_1\tilde{a})$ and every essential vertex $\tilde{s} \in \langle \tilde{a}\tilde{g}_1 \rangle$ satisfies $\det(\tilde{s}\tilde{g}_1) = 1$. The study of this case is completed analogously to 6), 7) in Fig. 12.

2) Assume that $\tilde{g}_1$ is incident to $D_{\tilde{a}}$. Then $a = h$. Applying [4], Lemma 3 to $\tilde{a}$, we get $\det L_a^1 = \det L_a = 3$, $\det L_a^3 = 1$. Analogously to Assertion 5.2, every nodal vertex $\tilde{s} \in (\tilde{a}\tilde{g}_2)$ satisfies $\det(\tilde{s}\tilde{g}_2) = 2$.

a) Suppose that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. By Lemma 3.15 and Corollary 3.8, we have (in the notation of Lemma 2.9) $\det\tilde{L}_\infty = -1$, $\mu_1 = \det(\tilde{a}\tilde{g}_2) = 1$, $d_1 = \det L_a^1 = 3$, $\mu_2 = \dots = \mu_n = 2$. This contradicts (2.6).

b) The impossibility of the case when $\tilde{v} \in (\tilde{a}\tilde{g}_1)$ is proved as in step 2) in the proof of Lemma 5.3.

§ 6. Prohibition of diagrams 3a), b) in Fig. 11

Assume that $\tilde{L}$ corresponds to one of the diagrams 3a) or b). We denote by $\tilde{a}$ (resp. $\tilde{b}$) the right (resp. left) vertex of the diagram.

Lemma 6.1. *The vertex $\tilde{g}_1$ is incident to $(\tilde{b}\tilde{g}_2)$.*

Proof. We work out the case 3a) in detail. The case 3b) is treated analogously. Assume the opposite. We denote by $D_{\tilde{a}}^i$ (resp. $D_{\tilde{b}}^i$) the connected component of $F^{-1}(D_a)$ (resp. $F^{-1}(D_b)$) of degree i incident to $\tilde{a}$ (resp. $\tilde{b}$) that corresponds to the lower branch of the diagram. We denote by $R_{\tilde{a}}^1$ (resp. $R_{\tilde{b}}^1$) the connected component of $F^{-1}(R_a)$ (resp. $F^{-1}(R_b)$) of degree 1 incident to $\tilde{a}$ (resp. $\tilde{b}$). Let L_1 and L_2 be the connected components of $F^{-1}(L_b)$. Then

$$\begin{aligned} \det D_{\tilde{a}}^1 &= \det D_a = 2, & \det D_{\tilde{a}}^2 &= 1, & \det R_{\tilde{a}}^1 &= \det(\tilde{a}\tilde{g}_2) = \det R_a = 1, \\ \det(\tilde{a}\tilde{b}) &= \frac{1}{3} \det(ab) > 0, & \det(\tilde{b}\tilde{g}_2) &= \det R_{\tilde{b}}^1 = \det R_b = 1, \\ \det D_{\tilde{b}}^1 &= 3 \det D_{\tilde{b}}^3, & \det L_1 &= \det L_2 = \frac{1}{2} \det L_b, \\ \det \tilde{L}_\infty &= -2 \det L_1 \det D_{\tilde{b}}^3 < 0. \end{aligned} \tag{6.1}$$

1) We claim that $\tilde{v} \in (\tilde{a}\tilde{g}_2) \cup (\tilde{a}\tilde{b})$. Indeed, if $\tilde{v} \notin \text{br}_{\tilde{a}}(\tilde{b})_\infty$, then $\det(\tilde{L}_\infty - \tilde{a}) > 0$, $\det \tilde{L}_\infty < 0$ and $\det(\tilde{a}\tilde{g}_2) = 1$ by Lemma 4 of [4] and (6.1). Hence Lemma 2.12 implies that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. If $\tilde{v} \in \text{br}_{\tilde{a}}(\tilde{b})_\infty$, then $\tilde{v} \in (\tilde{b}\tilde{a})$ as in step 1) in Lemma 5.10.

2) Suppose that $h = b$ and L is of type 1. Then Corollary 3.8 and Assertion 3.9 imply that $\det(\tilde{b}\tilde{g}_1) = 1$, $\det \delta(\tilde{b}\tilde{g}_1) = 1$. By Lemma 2.3, $\tilde{b}$ is inessential. This contradicts (6.1).

3) Let $\tilde{g}_1$ be incident to $R_{\tilde{b}}^1$. Since $\text{Deg } R_{\tilde{b}}^1 = 1$, the splice diagrams of $R_{\tilde{b}}^1$ and R_b are isomorphic. We see by Proposition 1.3, 2) and Lemma 2.3 that $\det \delta(\tilde{b}\tilde{g}_1) = 1$ and $\tilde{b}$ is inessential for $\tilde{g}_1$. This contradicts (6.1).

4) Suppose that $\tilde{g}_1$ is incident to L_1 or L_2 (say, L_1). By [4], Lemma 3 we have $\det D_{\tilde{b}}^1 = 3$, $\det D_{\tilde{b}}^3 = 1$, $\det L_1 = \det L_2 = 1$. Reasoning as in step 2) in Lemma 5.10, we get $\det \delta(\tilde{g}_1\tilde{b}) = 1$. Since $\det L_1 = 1$, the vertex $\tilde{b}$ is inessential by Lemma 2.3. This contradicts (6.1).

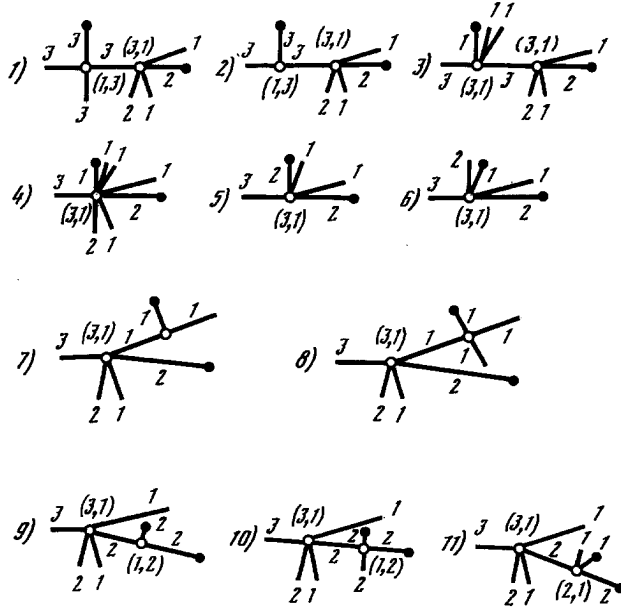
5) Suppose that $\tilde{g}_1$ is incident to $D_{\tilde{b}}^i$ for $i = 1$ or $i = 3$. Then $\det L_1 = \det L_2 = 1$ by Lemma 3 of [4] and (6.1). By Corollary 3.8, $\det \delta(\tilde{b}\tilde{g}_1) = 1$. Let $\tilde{s}$ be a nodal vertex. It is easy to prove that

$$\det(\tilde{s}\tilde{g}_2) = 3 \quad \text{if } \tilde{s} \in (\tilde{a}\tilde{b}), \quad \det(\tilde{s}\tilde{g}_2) = 2 \quad \text{if } \tilde{s} \in (\tilde{a}\tilde{g}_2). \tag{6.2}$$

a) Assume that $\tilde{v} \in (\tilde{a}\tilde{b})$. Combining (6.1), (6.2), Corollary 3.8 and Lemma 3.15, we have (in the notation of Lemma 2.11) $\det \tilde{L}_\infty = -2 \det D_{\tilde{b}}^3$, $\mu_2 = \dots = \mu_{n-1} = 3$, $\mu_n = 1$, $d_n = 2$. If $\tilde{b}$ is inessential, then $\mu_1 = 3$ and $\det \tilde{L}_\infty = -2$, which contradicts (2.12). If $\tilde{b}$ is essential, then $\mu_1 = 1$, and $d_1 = d(\tilde{b})$ equals either $\det D_{\tilde{b}}^1 = 3 \det D_{\tilde{b}}^3$ or $\det D_{\tilde{b}}^3$. Substituting these data in (2.12), we get $d_1 = \det D_{\tilde{b}}^3 - 4$, which is incompatible with either possibility for d_1 .

b) Assume that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. By Lemma 3.15, $(\tilde{v}\tilde{g}_2)$ is linear. According to (6.1), (6.5), there is an integer p such that (in the notation of Lemma 2.9) $\tilde{k}_p = \tilde{a}$, $\det \tilde{L}_\infty = -2 \det D_{\tilde{b}}^3$, $\mu_2 = \dots = \mu_{p-1} = 3$, $\mu_p = 1$, $d_p = 2$, $\mu_{p+1} = \dots = \mu_n = 2$.

We now study $\text{br}_b(\tilde{a})$ in detail. By Assertion 3.9 and Definition 3.10, $\text{br}_b(\tilde{a})$ corresponds to one of the diagrams in Fig. 13.



c) If $\tilde{L}$ corresponds to 11) in Fig. 13, then $\det D_a^1 = \det D_a = 2$, $\det D_a^2 = 1$, $\det(\tilde{a}\tilde{g}_2)_\infty = \det U$, $\det \tilde{L}_\infty = -2 \det U$, and the determinant of any subgraph that corresponds to one of the upper edges incident to $\tilde{h}$ is equal to $\det U$.

d) If $\tilde{L}_\infty$ corresponds to one of the diagrams 1)-4) in Fig. 13, and the nodal vertex $\tilde{s} \in \langle \tilde{h}\tilde{a} \rangle \cup \tilde{h}$ is not a fork, then $\det(\tilde{s}\tilde{g}_2) = 3$.

Lemma 6.4. *The case 1) in Fig. 13 is impossible.*

Proof. Put $D_{\tilde{h}} := F^{-1}(D_h) \cap (\tilde{a}\tilde{b})$. Then $D_{\tilde{h}}$ is linear, $\det D_{\tilde{h}} = \det D_h > 1$, $\det(\tilde{h}\tilde{g}_1) = \det U = 1$. By Lemmas 2.3 and 2.4, $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. Since $\tilde{h}$ is not a fork, Lemma 2.7 and Assertion 6.3, d) imply that $\det D_{\tilde{h}} = 1$, which is impossible.

Lemma 6.5. *If $\tilde{L}$ corresponds to one of the diagrams 2) or 3) in Fig. 13, then $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$.*

Proof. Assume that $\tilde{v} \notin (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. Then $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$, $\det \delta(\tilde{h}\tilde{g}_1) = 1$, the subgraphs $(\tilde{a}\tilde{g}_2)$ and $(\tilde{h}\tilde{a})$ are linear, and $\tilde{v} \in \text{br}_{\tilde{h}}(\tilde{b})$. Using Assertion 6.3 and applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $(\tilde{b}\tilde{g}_1)$, we obtain $\det(\tilde{b}\tilde{g}_1) = 1 - 2\det(\tilde{a}\tilde{b})_\infty < 0$. Thus $\tilde{v} \in (\tilde{h}\tilde{b})$. Note that $\det(\tilde{h}\tilde{g}_1) = \det U$. Using Proposition 1.3, 2) and Lemmas 2.3, 2.4, 2.7, we see that the following holds.

If $\det U = 1$, then $\det \text{br}_{\tilde{h}}(\tilde{g}_2) = 1$.

If $\det U \neq 1$, then $\tilde{L}$ corresponds to 2) in Fig. 13, $\det(\tilde{h}\tilde{g}_2) = 3$, $\det \text{br}_{\tilde{h}}(\tilde{g}_1) = 1$. (6.3)

As in step b) of Lemma 5.6, we prove that $(\tilde{v}\tilde{h})$ is linear.

If $\det U = 1$, then $\det \text{br}_{\tilde{g}_1}(\tilde{h}) = -2$ by [4], (3) and (6.3). Applying [4], (2) to $(\tilde{h}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{h})$ and to $(\tilde{a}\tilde{g}_2)$ as a subgraph of $\text{br}_{\tilde{h}}(\tilde{g}_2)$ and using Assertion 6.3 and (6.3), we get $-2\det(\tilde{h}\tilde{g}_2) = -2 - 4\det \text{br}_{\tilde{h}}(\tilde{b})$, $1 = \det(\tilde{h}\tilde{g}_2) - 2\det(\tilde{a}\tilde{h})$. Hence $\det \text{br}_{\tilde{h}}(\tilde{b}) = \det(\tilde{a}\tilde{h}) > 0$. Since $\det U = 1$, Lemma 2.12 implies that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. This contradicts the assumption that $\tilde{v} \notin (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$.

If $\det U \neq 1$, then $\det \text{br}_{\tilde{h}}(\tilde{g}_1) = 1$ by (6.3). Using formulae (2) and (3) of [4], we easily see that $\det \text{br}_{\tilde{g}_2}(\tilde{h}) = -7$. Combining (6.3), Assertion 6.3 and [4], (2) for $(\tilde{h}\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{g}_2}(\tilde{h})$, we get $-7\det U = -2 - 3\det \text{br}_{\tilde{h}}(\tilde{b})$. Thus $\det \text{br}_{\tilde{h}}(\tilde{b}) > \det U$, and Lemma 2.12 implies that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. This contradicts the assumption that $\tilde{v} \notin (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$.

Lemma 6.6. *The cases 2), 3) in Fig. 13 are impossible.*

Proof. Assume the opposite. By Assertion 6.3, $\det(\tilde{a}\tilde{g}_2) = 1$. We see from Lemma 2.12 and Lemma 6.5 that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{a}) \cup (\tilde{h}\tilde{g}_2)$. Let us prove that $\tilde{v}$ cannot belong to any of these three sets.

1) Suppose that $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. Then $(\tilde{a}\tilde{h})$ and $(\tilde{a}\tilde{g}_2)$ are linear, $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$. Formula (3) of [4] and Corollary 3.8 imply that $\det \text{br}_{\tilde{g}_1}(\tilde{h}) = -2$. Applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{h})$, we get $\det(\tilde{a}\tilde{g}_1) = 0$. If $\det U \neq 1$, then $\tilde{h}$ is inessential by Assertion 6.3, d) and Lemma 2.7. Thus we have (in the notation of Lemma 2.10) $\det \tilde{L}_\infty = -2$, $\mu_1 = \det(\tilde{a}\tilde{g}_1) = 0$, $d_1 = 2$, $\mu_2 = \dots = \mu_n = 1$ independently of the value of $\det U$. This contradicts (2.11).

2) Reasoning as in steps 3), 4) in the proof of Lemma 5.3, we get $\tilde{v} \notin (\tilde{h}\tilde{a}) \cup (\tilde{g}_2\tilde{a})$.

Lemma 6.7. *The case 4) in Fig. 13 is impossible.*

Proof. Assume the opposite. Combining Assertion 6.3, Lemma 2.3 and Lemma 2.4, we see that $\tilde{v} \in (\tilde{a}\tilde{g}_1)$. By Lemma 3.15, $(\tilde{a}\tilde{g}_2)$ is linear. In the notation of Lemma 2.10, we have $\mu_1 = \dots = \mu_n = 1$ and $\det \tilde{L}_\infty = -2$ by Corollary 3.8. This contradicts (2.11).

Lemma 6.8. *If $\tilde{L}_\infty$ corresponds to one of the diagrams 5), 6) in Fig. 13, then $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$.*

Proof. 1) We claim that $\tilde{v} \in \text{br}_{\tilde{a}}(\tilde{b}) \cup (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$. Indeed, $\det \text{br}_{\tilde{a}}(\tilde{b}) > 0$ holds for $\tilde{v} \notin \text{br}_{\tilde{a}}(\tilde{b})$. Using Definition 3.2, Proposition 1.3, 2) and Assertion 6.3, we get $\det(\tilde{L}_\infty - \tilde{a}) > 0$, $\det \tilde{L}_\infty < 0$, $\det(\tilde{a}\tilde{g}_2) = 1$. Hence Lemma 2.12 implies that $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$.

2) We claim that $\tilde{v} \notin \text{br}_{\tilde{a}}(\tilde{b})$. Assume the opposite. By Corollary 3.8 and Lemma 3.15, $(\tilde{a}\tilde{g}_2)$ is linear and $\det \delta(\tilde{a}\tilde{g}_1) = 1$.

The impossibility of the case when $\det \text{br}_{\tilde{a}}(\tilde{g}_1) = \det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$ is proved similarly to Lemma 5.7. (The determinants of the connected components of U' incident to $\tilde{a}$ are now different, but this does not influence our reasoning.)

Put $d = \max(\det \text{br}_{\tilde{a}}(\tilde{g}_1), \det \text{br}_{\tilde{a}}(\tilde{g}_2))$. Since $d > 1$, we see that $\tilde{L}$ corresponds to 6) in Fig.13, $\det U = 2 = \det(\tilde{a}\tilde{g}_1)$ and $\min(\det \text{br}_{\tilde{a}}(\tilde{g}_i)) = 1$, $i = 1, 2$. Assertion 6.3 implies that $\det \tilde{L}_\infty = -2$.

a) Suppose that $\det \text{br}_{\tilde{a}}(\tilde{g}_1) = d$. Applying [4], (2) to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\tilde{L}$ and to $(\tilde{a}\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{g}_2}(\tilde{a})$, we get $-1 = \det \text{br}_{\tilde{g}_2}(\tilde{a}) - d \det \text{br}_{\tilde{a}}(\tilde{b})$, $2 \det \text{br}_{\tilde{g}_2}(\tilde{a}) = -2d - \det \text{br}_{\tilde{a}}(\tilde{b})$. Hence $\det \text{br}_{\tilde{a}}(\tilde{b}) = -\frac{2d-2}{2d+1}$. But this number is not an integer since $d > 1$.

b) Suppose that $\det \text{br}_{\tilde{a}}(\tilde{g}_2) = d$. As above, we apply [4], (2) to $(\tilde{a}\tilde{g}_1)$ regarded as a subgraph of $\tilde{L}$ and to $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{a})$. We get $-2 = \det \text{br}_{\tilde{g}_1}(\tilde{a}) - d \det \text{br}_{\tilde{a}}(\tilde{b})$, $\det \text{br}_{\tilde{g}_1}(\tilde{a}) = -2d - 2 \det \text{br}_{\tilde{a}}(\tilde{b})$. Thus $\det \text{br}_{\tilde{a}}(\tilde{b}) = -\frac{2d-2}{d+2}$. Since $d > 1$, we have $d = 4$, $\det \text{br}_{\tilde{a}}(\tilde{b}) = -1$, $\det \text{br}_{\tilde{g}_1}(\tilde{a}) = -6$. Applying [4], (2) to $\tilde{L}_\infty$ regarded as a subgraph of $\tilde{L}$, we get $2 = -6 \det \text{br}_{\tilde{g}_2}(\tilde{a}) - 1$, which is impossible.

Lemma 6.9. *The cases 5), 6) in Fig. 13 are impossible.*

Proof. Assume the opposite. By Lemma 6.8, $\tilde{v} \in (\tilde{a}\tilde{g}_1) \cup (\tilde{a}\tilde{g}_2)$.

1) Suppose that $\tilde{v} \in (\tilde{a}\tilde{g}_2)$. If $\tilde{a}$ is inessential, then $\tilde{L}$ corresponds to 6), $\det U = 2$ and we have (in the notation of Lemma 2.9) $\mu_1 = \dots = \mu_n = 2$ and $\det \tilde{L}_\infty = -2$, which contradicts (2.5). If $\tilde{a}$ is essential, then (in the notation of Lemma 2.9) $\mu_1 = 1, \mu_2 = \dots = \mu_n = 2$, $\det \tilde{L}_\infty = -\det U$ and $d_1 = d(\tilde{a})$ equals to one of the numbers $\det U$, $\frac{1}{2} \det U$ and $\det \text{br}_{\tilde{a}}(\tilde{b})$. The equality $d_1 = \det \text{br}_{\tilde{a}}(\tilde{b})$ holds only if $\tilde{L}$ corresponds to 6) and $\det U = 2$. Substituting these data in (2.6), we get $0 = 4 - \det U + d_1$. This is possible only if $\det U = 8$ and $d(\tilde{a}) = 4$. By [4], (3) we have $\det \text{br}_{\tilde{g}_2}(\tilde{v}) = -3$. Applying Lemma 3 of [4] to $\tilde{a}$, we have $\det \text{br}_{\tilde{a}}(\tilde{b}) = \det \text{br}_{\tilde{a}}(\tilde{g}_1) = 1$. Combining [4], (2) for $(\tilde{a}\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{g}_2}(\tilde{v})$ with Assertion 6.3, we obtain $-3 \det U = -\det U - d(\tilde{a})$, which contradicts the equations $\det U = 8$, $d(\tilde{a}) = 4$.

2) Suppose that $\tilde{v} \in (\tilde{a}\tilde{g}_1)$. If $\tilde{a}$ is inessential, then we obtain a contradiction as in step 2) in the proof of Lemma 5.4. If $\tilde{a}$ is essential, then $\det \text{br}_{\tilde{g}_1}(\tilde{a}) = -2$,

$\det \text{br}_{\tilde{a}}(\tilde{g}_2) = 1$ as in step 2) in Lemma 5.3. Combining Assertion 6.3 with [4], (2) for $(\tilde{a}\tilde{g}_2)$ regarded as a subgraph of $\text{br}_{\tilde{g}_1}(\tilde{a})$, we obtain $-2 = -\det U - \frac{1}{2} \det {}^2U \det \text{br}_{\tilde{a}}(\tilde{b})$. This contradicts the equation $\det U \geq 2$.

The following lemma is proved analogously to Lemmas 5.5 and 5.8.

Lemma 6.10. *The cases 7), 8), 10) in Fig. 13 are impossible.*

Lemma 6.11. *The cases 9), 11) in Fig. 13 are impossible.*

Proof. Assume the opposite.

1) We claim that $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. The case 11) with $\det U \neq 1$ is considered as in step 1) in the proof of Lemma 5.7. Let us study the other cases: 11) with $\det U = 1$ and 9). By Assertion 6.3, $\det \tilde{L}_\infty = -2$. Assume that $\tilde{v} \notin (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$. As in steps a), b) in the proof of Lemma 5.6, we see that $\tilde{v} \in [\tilde{h}\tilde{a}]$, and $(\tilde{v}\tilde{h})$ is linear. In analogy with step c) in Lemma 5.6 (where $\det \tilde{L}_\infty = -1$, while now $\det \tilde{L}_\infty = -2$), we obtain

$$\det \text{br}_{\tilde{h}}(\tilde{a}) = \begin{cases} 2 \det(\tilde{h}\tilde{g}_2) - 2 \geq 0 & \text{if } \det U = 1, \\ \frac{3 \det U - 2}{2} \geq 2 & \text{if } \det U \neq 1. \end{cases} \quad (6.4)$$

If $\det \text{br}_{\tilde{h}}(\tilde{a}) = 0$, then $\det(\tilde{h}\tilde{g}_2) = 1$, which contradicts Lemma 2.8. Using (6.4) and Lemma 2.12, we get $\tilde{v} \in (\tilde{h}\tilde{g}_1) \cup (\tilde{h}\tilde{g}_2)$, which contradicts our assumption.

2) The impossibility of the cases 11) with $\det U = 1$ and 9) can be proved analogously to steps 2), 3) in Lemma 5.6. We claim that the case 11) with $\det U \neq 1$ is also impossible.

a) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_2)$. In the notation of Lemma 2.9, we have $\mu_1 = 1$, $d_1 = \det U$, $\det \tilde{L}_\infty = -2 \det U$, $\mu_2 = \dots = \mu_n = 2$. Substituting these data in (2.6), we get $\det U = 4$. By [4], Lemma 5, $\det(ah) = 2 \det(\tilde{a}\tilde{h})$. Hence the integers $\det U$ and $\det(ah)$ are not coprime, which contradicts the equation $\det R_a = 1$.

b) Assume that $\tilde{v} \in (\tilde{h}\tilde{g}_1)$. Analogously to step 3) in Lemma 5.7 (where we had $\det \tilde{L}_\infty = -\det U$, while now $\det \tilde{L}_\infty = -2 \det U$), we get $\det U = 2$, which is impossible for the same reasons as in a).

Combining Lemmas 6.4, 6.6, 6.7, 6.9–6.11, we obtain the following lemma.

Lemma 6.12. *The cases 3a), 3b) in Fig. 11 are impossible.*

§ 7. Prohibition of diagrams 4a), b) in Fig. 11

We study the case 4b) in detail. (The case 4a) is similar.) We denote by $\tilde{a}_1$ (resp. $\tilde{b}_2$) the fork corresponding to the left (resp. right) vertex. We put $a := F(\tilde{a}_1)$, $b := F(\tilde{b}_2)$, $\tilde{a}_2 := a' \setminus \tilde{a}_1$, $\tilde{b}_1 := b' \setminus \tilde{b}_2$. If $a = h$ or $b = h$, then L is of type 1 or 2.

The upper (resp. lower) diagram in Fig. 14 corresponds to $\tilde{L}_\infty + \tilde{g}_2$ (resp. L). The subgraphs (ab) , R_b and L_a of the graph L are depicted by edges. The edge below a (resp. b) corresponds to D_a (resp. D_b). (We recall that $D_h = U$ if L is of type 2; see the convention in § 1.) The edges of the upper diagram correspond to the connected components of $F^{-1}(L_a \cup (ab) \cup R_b \cup D_a \cup D_b)$. The numbers on the edges correspond to the determinants of the subgraphs (see the details below).

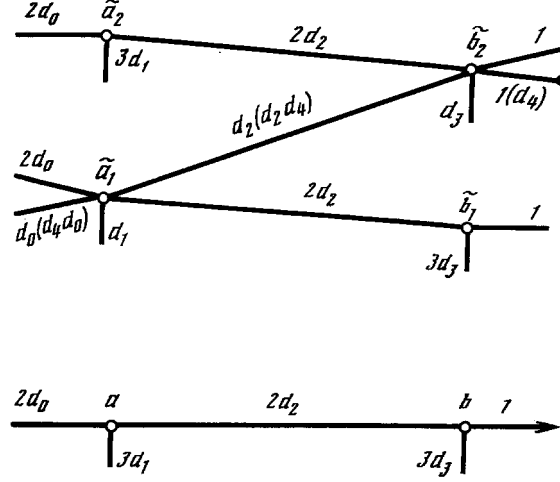


Figure 14

Hence, if L is of type 1 and h coincides with a or b , then all connected components of $L - a - b$ except for U (and all connected components of $F^{-1}(L - a - b)$ except those contained in U') are depicted in Fig. 14. The non-depicted components will not influence what follows (see § 3).

We need some auxiliary calculations. By Lemma 3.14, all edges in 4b) except possibly for one correspond to “edge type” subgraphs. This exceptional edge corresponds to the subgraph $\tilde{V}$ of the form a), c) in Fig. 8 incident to $\tilde{g}_1$. Let d_4 be the determinant of the branch of degree 1 in $\tilde{V}$. By Definition 3.2 and [4], (6), we get $\det \tilde{V} = d_4 \frac{n(\tilde{p})n(\tilde{q})}{2} \det F(\tilde{V})$, where $\tilde{p}, \tilde{q}$ are vertices of $\tilde{L}_\infty - \tilde{g}_1$ incident to $\tilde{V}$. (We put $n(\emptyset) = 1$.) The numbers on the edges are the determinants of the corresponding subgraphs. The number in the brackets corresponds to $\tilde{V}$.

We see from Fig. 14 that $d_1, d_3, 2d_0, 2d_2$ are integers. We claim that d_0 is an integer. According to Fig. 14, it suffices to prove that d_4 is odd if $\tilde{V}$ is the left branch of $\tilde{L}_\infty$ incident to $\tilde{a}_1$. Suppose that $\tilde{V}$ corresponds to a) in Fig. 8. (The case c) in Fig. 8 is treated analogously.) By Definition 3.2, $\det U = d_4$ and $\det L_h$ is even. Hence Lemma 2 of [4] implies that d_4 is odd. Since $n(\tilde{a}_1) = 1$ and $\det R_a = 1$, reasoning as before we obtain that d_2 is an integer.

We omit some intermediate calculations, which are similar to those in [4], Lemma 10. Put $\delta := \det \delta(ab)$. Using [4], Lemma 5 and Definition 3.2, we get

$$\det \delta(\tilde{a}_2 \tilde{b}_2)_\infty = \det \delta(\tilde{a}_1 \tilde{b}_1)_\infty = \delta, \quad \det \delta(\tilde{a}_1 \tilde{b}_2)_\infty = \lambda \delta. \quad (7.1)$$

Here and in what follows we put $\lambda = 1$ if $\delta(a_1 b_2)_\infty$ does not contain $\tilde{V}$, and $\lambda = d_4$ otherwise. Using Definition 3.2 and [4], (2) for (ab) regarded as a subgraph of L , we find

$$\det \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty = \det L_b = 18d_0 d_1 d_3 \delta^2 - 2d_2, \quad \det \text{br}_{\tilde{a}_1}(\tilde{b}_1)_\infty = \det R_a = 1. \quad (7.2)$$

We denote by $\Delta_{\tilde{a}_1}$ (resp. $\Delta_{\tilde{b}_2}$) the connected component of $F^{-1}(L_b)$ (resp. $F^{-1}(R_a)$) that contains $\tilde{a}_1$ (resp. $\tilde{b}_2$). Applying [4], (6) to $\tilde{a}_1 \in \Delta_{\tilde{a}_1}$ and $\tilde{b}_2 \in \Delta_{\tilde{b}_2}$, we get

$$\det \Delta_{\tilde{a}_1} = \lambda d_0 d_2 \det L_b, \quad \det \Delta_{\tilde{b}_2} = \lambda d_2. \quad (7.3)$$

Combining (7.1)–(7.3) with formula [4], (2) for $(\tilde{a}_1 \tilde{b}_1)_\infty$ regarded as a subgraph of $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty$, we get $2d_2 \det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty = \lambda(d_0 d_2 \det L_b - 6d_0^2 d_1 d_2 d_3 \delta^2)$. Substituting this in (7.2), we obtain

$$\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty = \lambda d_0 (6d_0 d_1 d_3 \delta^2 - d_2). \quad (7.4)$$

Analogously, combining (7.1)–(7.3) with formula [4], (2) for $(\tilde{a}_2 \tilde{b}_2)_\infty$ regarded as a subgraph of $\text{br}_{\tilde{a}_1}(\tilde{b}_2)_\infty$, we find

$$\det \text{br}_{\tilde{a}_1}(\tilde{b}_2)_\infty = \lambda (6d_0 d_1 d_3 \delta^2 - d_2). \quad (7.5)$$

We claim that

$$\det \tilde{L}_\infty = \lambda d_0 (-8d_0 d_1 d_3 \delta^2 + d_2), \quad \det \text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty = \lambda d_0 (2d_0 d_1 d_3 \delta^2 - d_2). \quad (7.6)$$

Indeed, let us study the case when $(\tilde{a}_1 \tilde{b}_2) = \tilde{V}$. The other cases are studied analogously. Substituting (7.1), (7.4), (7.5) in formula [4], (2) for $(\tilde{a}_1 \tilde{b}_2)_\infty$ regarded as a subgraph of $\tilde{L}_\infty$, we get

$$d_4 d_2 \det \tilde{L}_\infty = (6d_0 d_1 d_3 \delta^2 - d_2)^2 d_4^2 d_0 - d_3 \det L_b 2d_0^2 d_1 d_3 \delta^2 d_4^2.$$

Using (7.2), we obtain the first formula in (7.6). Substituting (7.1)–(7.4) in formula [4], (2) for $(\tilde{a}_1 \tilde{b}_2)_\infty$ regarded as a subgraph of $\text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty$, we get

$$d_4 d_2 \det \text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty = d_0 d_2 d_4^2 (6d_0 d_1 d_3 \delta^2 - d_2) - 4d_2 d_0^2 d_3 d_1 d_4^2 \delta^2.$$

Hence the second formula in (7.6) is valid.

Put $L_{\tilde{a}_2} = F^{-1}(L_a) \cap \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$, $D_{\tilde{a}_2} = F^{-1}(D_a) \cap \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$.

Lemma 7.1. 1) If $\tilde{g}_1$ is not incident to $L_{\tilde{a}_2} \cup D_{\tilde{a}_2}$, then $\tilde{v} \in L_{\tilde{a}_2} \cup D_{\tilde{a}_2}$.

2) If $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty \leq 0$, then $\tilde{v} \in (\tilde{a}_1 \tilde{b}_2)_\infty$.

3) If $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty > 0$, then $\det \tilde{L}_\infty < 0$ and $\tilde{v} \in (\tilde{b}_2 \tilde{g}_2) \cup \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$.

Proof. 1) $L_{\tilde{a}_2}$ and $D_{\tilde{a}_2}$ are branches of $\tilde{L}$. By Fig. 14, $\min(\det L_{\tilde{a}_2}, \det D_{\tilde{a}_2}) > 1$. Hence the statement follows from [4], Lemma 3.

2) By (7.4) and (7.5), $\det \text{br}_{\tilde{a}_1}(\tilde{b}_2)_\infty \leq 0$. Hence $\tilde{v} \in \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty \cap \text{br}_{\tilde{a}_1}(\tilde{b}_2)_\infty = (\tilde{a}_1 \tilde{b}_2)_\infty$.

3) By (7.4), $d_2 < 6d_0 d_1 d_3 \delta^2$. Substituting this in (7.6), we get $\det \tilde{L}_\infty < -2d_0 d_1 d_3 \delta^2 < 0$. Assume that $\tilde{v} \notin (\tilde{b}_2 \tilde{g}_2) \cup \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$. Then $\tilde{g}_1 \in \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$ by 1). Moreover, $\det(\tilde{b}_2 \tilde{g}_2) = 1$ and $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty > 0$. Using Lemma 2.12, we see that $\tilde{v} \in (\tilde{b}_2 \tilde{g}_2) \cup \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$. This contradicts our assumption.

Lemma 7.2. *If $\tilde{v} \in L_{\tilde{a}_2} \cup D_{\tilde{a}_2}$, then $\tilde{g}_1 \in (\tilde{b}_2\tilde{g}_2) \cup \text{br}_{\tilde{b}_2}(\tilde{a}_2)$.*

Proof. Since $\tilde{v} \in \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty$, it follows that $\det(\tilde{L}_\infty - \tilde{b}_2 - \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty) > 0$. Hence $\det \tilde{L}_\infty < 0$ by Lemma 7.1 3). By (7.2) and Proposition 1.3, 2), $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2)_\infty = \det L_b > 0$. Thus $\det(\tilde{L}_\infty - \tilde{b}_2) > 0$.

If $\tilde{g}_1 \notin (\tilde{b}_2\tilde{g}_2)$, then $\det(\tilde{b}_2\tilde{g}_2) = 1$ and Lemma 2.12 implies that $\tilde{v} \in (\tilde{b}_2\tilde{g}_2) \cup \text{br}_{\tilde{b}_2}(\tilde{g}_1)$. Using the hypothesis of the lemma, we get $\tilde{g}_1 \in \text{br}_{\tilde{b}_2}(\tilde{a}_2)$.

Lemma 7.3. $\tilde{v} \notin L_{\tilde{a}_2} \cup D_{\tilde{a}_2}$.

Proof. Assume the opposite. By Lemma 4 of [4], $\det \text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty > 0$. By Lemma 7.2, $\text{br}_{\tilde{b}_2}(\tilde{a}_1) = \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty$. Combining (7.4), (7.6) and [4], Lemma 3, we get

$$\det \text{br}_{\tilde{b}_2}(\tilde{a}_1) > 4, \quad \det \text{br}_{\tilde{b}_2}(\tilde{g}_2) = d_3 = d_0 = d_1 = 1. \quad (7.7)$$

1) Suppose that $\tilde{g}_1 \in \text{br}_{\tilde{b}_2}(\tilde{a}_2)$. Then $\det(\tilde{b}_2\tilde{g}_2) = 1$, $\det(\tilde{a}_1\tilde{b}_2) = d_2$. Put $\Delta := \Delta_{\tilde{b}_2} \cup \tilde{g}_2$. Since $\tilde{v} \notin \Delta$, we have $\det \Delta > 0$. Combining (7.3) and (7.7) with [4], (2) for $(\tilde{b}_2\tilde{g}_2)$ regarded as a subgraph of Δ , we get $\det \Delta = d_2 - 2d_2^2 \det^2 \delta(\tilde{b}_2\tilde{g}_2) < 0$, a contradiction.

2) Suppose that $\tilde{g}_1 \in (\tilde{b}_2\tilde{g}_2)$. By Lemma 3 of [4], $\det \text{br}_{\tilde{a}_2}(\tilde{g}_2) = 1$. Combining (7.1), (7.2), (7.7) with [4], (2) for $(\tilde{b}_2\tilde{a}_2)$ regarded as a subgraph of $\tilde{L}$, we get $-2d_2 = (18\delta^2 - 2d_2) - 6 \det \text{br}_{\tilde{b}_2}(\tilde{a}_1)\delta^2$. Thus $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1) = 3$, which contradicts (7.7). We obtain that $\tilde{g}_1 \notin \text{br}_{\tilde{b}_2}(\tilde{a}_2) \cup (\tilde{b}_2\tilde{g}_2)$. This contradicts Lemma 7.2 and the assumption.

Corollary 7.4. $\tilde{g}_1$ is incident to $L_{\tilde{a}_2} \cup D_{\tilde{a}_2}$, $\tilde{v} \in (\tilde{a}_1\tilde{b}_2) \cup (\tilde{b}_2\tilde{g}_2) \cup (\tilde{b}_2\tilde{a}_2)$, and $d_0 = d_1 = 1$.

Proof. This follows from Lemma 7.1, Lemma 7.3 and [4], Lemma 3 applied to $\tilde{a}_1$ and $\tilde{a}_2$.

Thus we have proved that all edges in Fig. 14 correspond to “edge type” subgraphs.

Lemma 7.5. $\det \text{br}_{\tilde{a}_2}(\tilde{g}_1) = 1$, $\det \delta(\tilde{g}_1\tilde{a}_2) = 1$.

Proof. The first statement follows from Corollary 7.4 and [4], Lemma 3 applied to $\tilde{a}_2$. The second statement is proved analogously to step 2) in Lemma 5.10.

Lemma 7.6. *If $\tilde{g}_1$ is incident to $L_{\tilde{a}_2}$, then $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2) = 9d_3\delta^2 - 4d_2$. If $\tilde{g}_1$ is incident to $D_{\tilde{a}_2}$, then $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2) = 6d_3\delta^2 - 2d_2$,*

Proof. This follows from (7.2), Corollary 7.4, Lemma 7.5 and [4], (2) applied to $(\tilde{a}_2\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{b}_2}(\tilde{g}_1)$.

Lemma 7.7. $\tilde{v} \in (\tilde{b}_2\tilde{a}_2) \cup (\tilde{b}_2\tilde{g}_2)$.

Proof. Assume the opposite. By Corollary 7.4 and Lemma 7.1, 3), $\tilde{v} \in (\tilde{a}_1\tilde{b}_2)$ and $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)_\infty \leq 0$. This together with (7.4) implies that $d_2 \geq 6d_3\delta^2$. Thus $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2) < 0$ by Lemma 7.6. This contradicts the fact that $\tilde{v} \in (\tilde{a}_1\tilde{b}_2)$.

Lemma 7.8. $\delta(ab) = \emptyset$.

Proof. Assume the opposite. Then there is a nodal vertex $k \in (ab)$ such that (ka) is linear. Put $\tilde{k}_1 := (\tilde{a}_1\tilde{b}_1) \cap k'$, $\tilde{k}_2 := (\tilde{a}_1\tilde{b}_2) \cap k'$, $\tilde{k}_3 := (\tilde{a}_2\tilde{b}_2) \cap k'$, $d_5 := \det(ak)$, $d_6 := \det D_k$, $d_7 = \det(kb)$. Since $(\tilde{a}_2\tilde{b}_2)$, $(\tilde{a}_1\tilde{b}_2)$ and $(\tilde{a}_1\tilde{b}_1)$ are subgraphs of constant degree, the vertices $\tilde{k}_1$, $\tilde{k}_2$, $\tilde{k}_3$ are not forks. Consider Fig. 15, which contains more detail than Fig. 14. We label each edge with the determinant of the corresponding subgraph. These are calculated using [4], Lemma 5.

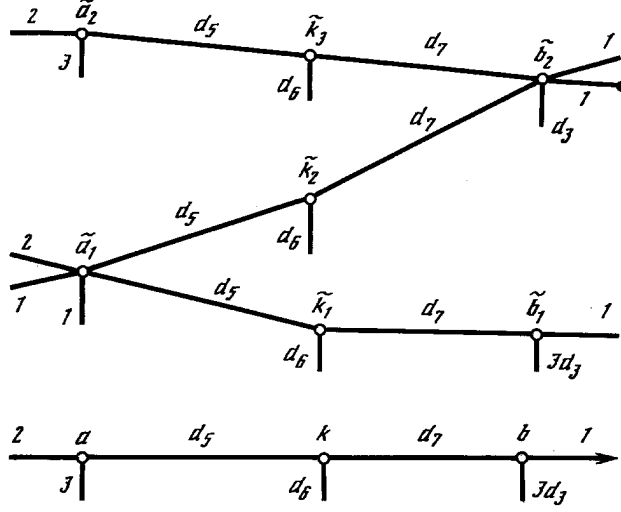


Figure 15

We claim that $\tilde{v} \in \text{br}_{\tilde{k}_3}(\tilde{a}_2)$. Indeed, using Proposition 1.3, 2), we get $\det R_k = 1$, $\det R_a = 1$, $d_6 > 1$. Analogously to the proof of (7.2), (7.3), we get

$$\begin{aligned} \det(\tilde{k}_3\tilde{g}_1) &= \det L_k, & \det(\tilde{k}_1\tilde{k}_2) &= d_5 \det L_k, \\ \det(\tilde{k}_3\tilde{k}_2)_\infty &= d_7, & \det L_k &= 6d_6 - d_5. \end{aligned} \quad (7.8)$$

Applying [4], Lemma 3 to the vertices $\tilde{k}_2$, $\tilde{a}_1$, $\tilde{k}_1$, we find that

$$\det \text{br}_{\tilde{k}_2}(\tilde{a}_1) = \det \text{br}_{\tilde{a}_1}(\tilde{b}_1) = \det \text{br}_{\tilde{k}_1}(\tilde{b}_1) = 1. \quad (7.9)$$

Combining (7.8), (7.9) with [4], (2) applied to $(\tilde{a}_1\tilde{k}_1)$ regarded as a subgraph of $\text{br}_{\tilde{k}_2}(\tilde{a}_1)$, we get

$$\det L_k = 1 + 2d_6, \quad d_5 = 4d_6 - 1. \quad (7.10)$$

Let us determine the dependence of $\det \text{br}_{\tilde{k}_3}(\tilde{a}_2)$ on the position of $\tilde{g}_1$. Combining Lemma 7.5 with [4], (2) for $(\tilde{a}_2\tilde{g}_1)$ regarded as a subgraph of $\text{br}_{\tilde{k}_3}(\tilde{a}_2)$, we get

$$\det \text{br}_{\tilde{k}_3}(\tilde{a}_2) = \begin{cases} 2 - 5d_6 < 0 & \text{if } \tilde{g}_1 \text{ is incident to } L_{\tilde{a}_2}, \\ 1 - 2d_6 < 0 & \text{if } \tilde{g}_1 \text{ is incident to } D_{\tilde{a}_2}. \end{cases}$$

Hence $\tilde{v} \in \text{br}_{\tilde{k}_3}(\tilde{a}_2)$. By Lemma 3.15, $\det \delta(\tilde{b}_2 \tilde{g}_2) = 1$. Combining (7.8) with [4], (2) applied to $(\tilde{b}_2 \tilde{g}_2)$ regarded as a subgraph of $(\tilde{k}_3 \tilde{k}_2)$, we get

$$\det(\tilde{k}_3 \tilde{k}_2) = d_7 \det \text{br}_{\tilde{b}_2}(\tilde{g}_2) - d_7^2 d_3.$$

Since $\tilde{v} \notin (\tilde{k}_3 \tilde{k}_2)$, we have $\det(\tilde{k}_3 \tilde{k}_2) > 0$. Thus $\det \text{br}_{\tilde{b}_2}(\tilde{g}_2) > 1$. Applying [4], Lemma 3 to $\tilde{b}_2$, we obtain that $d_3 = \det \text{br}_{\tilde{b}_2}(\tilde{a}_1) = 1$. Combining this with (7.8), (7.9) and [4], (2) applied to $(\tilde{k}_2 \tilde{b}_2)$ regarded as a subgraph of $(\tilde{k}_3 \tilde{g}_2)$, we get $d_7 \det(\tilde{k}_3 \tilde{g}_2) = d_7 - d_7 d_6 \det^2 \delta(\tilde{k}_2 \tilde{b}_2) < 0$. Since $\tilde{v} \in \text{br}_{\tilde{k}_3}(\tilde{a}_2)$, this contradicts [4], Lemma 4.

By Lemma 7.7, to prohibit the case 4b) in Fig. 11, it suffices to prove that $\tilde{v} \notin (\tilde{b}_2 \tilde{g}_2) \cup (\tilde{a}_2 \tilde{b}_2)$.

Lemma 7.9. $\tilde{v} \notin (\tilde{b}_2 \tilde{g}_2)$.

Proof. Assume the opposite. Then we have $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2) > 0$. By Lemma 7.6 and (7.4), $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1) > 1$. Applying [4], Lemma 3 to $\tilde{b}_2$, we get $\det \text{br}_{\tilde{b}_2}(\tilde{a}_2) = 1, d_3 = 1$. By substituting this in Lemma 7.6, (7.4) and (7.6) and using Lemma 7.8, we find that

$$\begin{aligned} \tilde{g}_1 \text{ is incident to } L_{\tilde{a}_2}, \quad d_2 = 2, \quad \det \text{br}_{\tilde{b}_2}(\tilde{a}_1) = 4, \\ \det \tilde{L}_\infty = -6, \quad \det \text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty = 0. \end{aligned} \quad (7.11)$$

Combining Lemma 3.15, Lemma 7.5 and (7.11), we have (in the notation of Lemma 2.9) $\mu_1 = \det(\tilde{a}_2 \tilde{g}_2) = 0$, $d_1 = \det D_{\tilde{a}_2} = 3$, $\mu_2 = \det(\tilde{b}_2 \tilde{g}_2) = 1$, $d_2 = \det \text{br}_{\tilde{b}_2}(\tilde{a}_1) = 4$, $\mu_3 = \dots \mu_n = 2$, $\det \tilde{L}_\infty = -6$, which contradicts (2.5).

Lemma 7.10. $\tilde{v} \notin (\tilde{a}_2 \tilde{b}_2)$.

Proof. Assume the opposite. By Lemmas 3.15 and 7.5, $\det \delta(\tilde{b}_2 \tilde{g}_2) = \det \delta(\tilde{g}_1 \tilde{a}_2) = 1$. By [4], Lemma 3, two of the three integers $\det \text{br}_{\tilde{b}_2}(\tilde{g}_2)$, d_3 , $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1)$ must be equal to one.

1) Suppose that $\tilde{b}_2$ is inessential, that is. $d_3 = 1$, $\det \text{br}_{\tilde{b}_2}(\tilde{a}_1) = 1$. We see from (7.4) and (7.6) that $d_2 = 5$, $\det \tilde{L}_\infty = -3$, $\det \text{br}_{\tilde{a}_2}(\tilde{b}_2)_\infty = -3$. In the notation of Lemma 2.9, we have $n = 1$, $\mu_1 = -3$, and $d_1 = 2$ or 3 . Substituting these data in (2.5), we get $0 = 5d_1 - 3 > 0$, a contradiction. (Here $d_1 := d(\tilde{a}_2)$.)

2) Suppose that $\tilde{b}_2$ is essential. Put $\delta_1 = d(\tilde{a}_2)$, $\delta_2 = d(\tilde{b}_2)$, $\mu_1 = \det(\tilde{a}_2 \tilde{g}_2)$. Then δ_1 equals 2 or 3, and $\delta_2 = \max(d_3, 6d_3 - d_2)$. Since $\det(\tilde{b}_2 \tilde{g}_2) = 1$, it follows from Lemma 2.11 that

$$0 = 1 + \det \tilde{L}_\infty + (\delta_1 + \delta_2)(\delta_2 + 1) - \mu_1(\delta_1 - 1). \quad (7.12)$$

Here $\det \tilde{L}_\infty = -8d_3 + d_2$, $\mu_1 = 2d_3 - d_2$ by (7.6). Successive consideration of all cases shows that neither of the possibilities for δ_1, δ_2 satisfies (7.12).

Combining Lemmas 7.7, 7.9 and 7.10, we obtain the following lemma.

Lemma 7.11. *The cases 4a), b) in Fig. 11 are impossible.*

We have proved that none of the diagrams of Fig. 11 is realized. This proves the main theorem.

Bibliography

- [1] A. G. Vitushkin, “On polynomial transformation of $\mathbb{C}^n$ ”, Manifolds, Tokyo, 1973, Tokyo Univ. Press, Tokyo 1975, pp. 415–417.
- [2] H. Bass, E. H. Connell, and D. Wright, “The Jacobian conjecture: Reduction of degree and formal expansion of the inverse”, *Bull. Amer. Math. Soc.* **7** (1982), 287–330.
- [3] S. Yu. Orevkov, “On three-sheeted polynomial mappings of $\mathbb{C}^2$ ”, *Izv. Akad. Nauk SSSR Ser. Mat.* **50** (1986), 1231–1240; English transl., *Math. USSR-Izv.* **29** (1987), 587–596.
- [4] A. V. Domrina and S. Yu. Orevkov, “On four-sheeted polynomial mappings of $\mathbb{C}^2$. I. The case of an irreducible ramification curve”, *Mat. Zametki* **64** (1998), 847–862; English transl. in *Math. Notes* **64** (1998).

Received 5/MAR/98

Translated by THE AUTHOR